CSE 4615:: Wireless Networks

IEEE 802.11 Wireless Local Area Networks

Lecture 6.1

Course Teacher:

Ashraful Alam Khan
Assistant Professor, CSE, IUT

Content Contributors:

Bernhard H. Walke
ComNets, RWTH Aachen University, Germany
Stefan Mangold
Swisscom Innovations, Switzerland
Lars Berlemann
ComNets, RWTH Aachen University, Germany

IEEE 802.11 Wireless Local Area Networks

Chapter 5, IEEE 802 Wireless Systems

Contents

- Scope of 802.11
- Reference Model, Architecture, Services, Frame Formats
- Physical Layer
- Medium Access Control Protocol
- Medium Access Control with Support for Quality- of-Service
- Radio Spectrum Management
- History and Selected Sub-standards, i.e., Amendments

SCOPE OF 802.11

Scope of 802.11

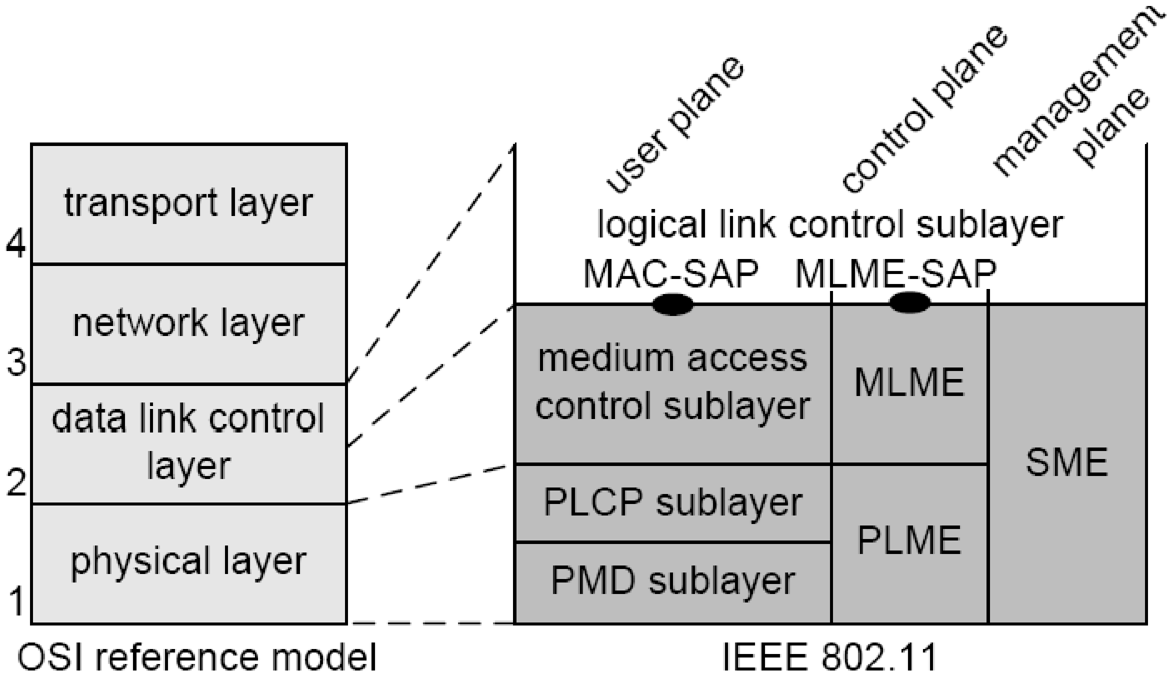
- 802.11 standard focuses on the 2 lower layers (1 & 2) of the OSI reference model.
- In the reference model of 802.11, Data Link Control (DLC) layer is divided into
  - Logical Link Control (LLC) and
  - Medium Access Control (MAC) sublayers.
- 802.11 defines Physical layer (PHY) transmission schemes and the MAC protocol, but no LLC functionality.

REFERENCE MODELS, ARCHITECTURE, SERVICES, FRAME FORMATS

Reference Model

- It is an abstraction of the real-life hardware and software modules of a radio standard, helping developers to discuss and understand the technology.

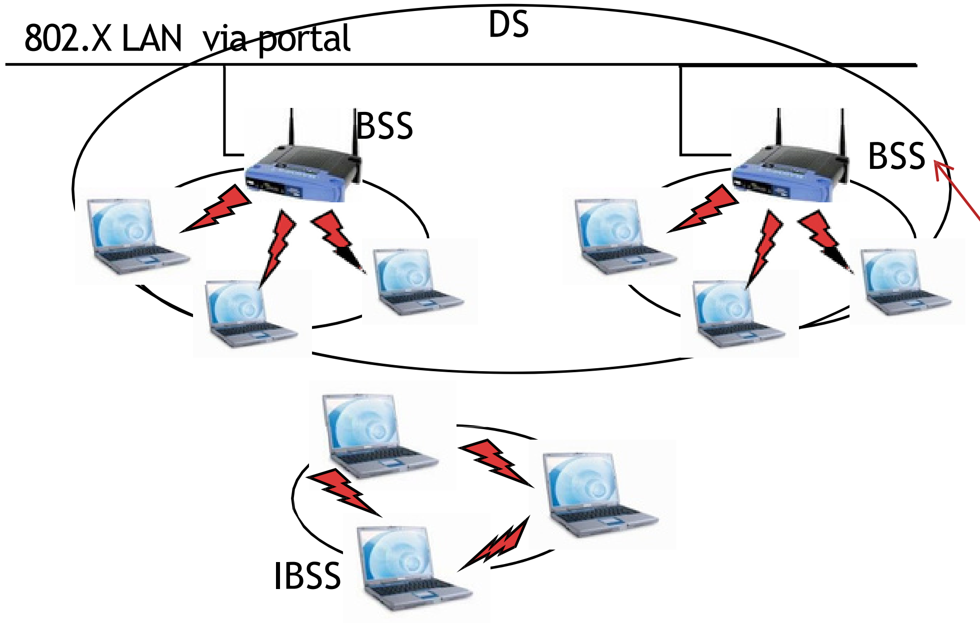
Reference Model



- Layer 1: PHY transmission scheme
- Layer 2: MAC protocols, MAC Layer Management Entity (MLME)
- Layer 1 & 2: Station Management Entity (SME)
- Interface to higher layers: Service Access Points (SAPs)

Architecture

802.11 Architecture & Coordination Functions



Basic Service Set (BSS):
- The basic element of IEEE 802.11 Networks
- A group of stations controlled by so-called Coordination Function (CF)

Coordination Function (CF):
- Manages the access to the wireless medium

Architecture

- Coordination Function (CF)
  1. The Distributed CF (DCF) is used by all stations in the BSS,
  2. whereas the Point CF (PCF) is an optional extension for the support of QoS.

  - DCF and PCF are concepts for spectrum management and medium access.

- Independent Basic Service Set (IBSS)
  - The simplest 802.11 network type. Consists of a minimum of two stations
  - No station has priority over another.
  - The responsibility of coordinating the medium access is distributed among all stations

Architecture

- An infrastructure-based BSS includes 1 station that has access to the wired network (Access Point-AP).
- An ESS (Extended Service Set) consists of 1 or more BSSs connected over the Distribution System (DS).

Services

- DS provides the service to transport MAC Service Data Units (MSDUs) between stations that are not in direct communication
- An AP provides the Distribution System Service (DSS).
  - The DSSs enable the MAC to transport MSDUs between stations that are not in direct communication.
  - So DSSs are not available in an IBSS.
- There are 2 categories of services in 802.11:
  - the Station Service (SS) and
  - the Distribution System Service (DSS).

Services

- The main SS of a BSS is
  - the MAC Service Data Unit (MSDU) delivery.
  - Other SSs include (de)-authentication and privacy.
- DSSs include
  - (re-)association, (dis)-association, and integration.
  - The integration enables the delivery of MSDUs between non-802.11 LANs and the DS via the so- called portal.

IEEE 802.11 Wireless LANs

802.11b

- **2.4-2.5 GHz unlicensed spectrum**
- **up to 11 Mbps**

802.11a

- 5-6 GHz range
- up to 54 Mbps
- OFDM

802.11g

- 2.4-2.5 GHz range
- up to 54 Mbps
- OFDM

802.11n: multiple antennae

- 2.4-2.5 and 5-6 GHz range
- up to 600 Mbps
  - MIMO
  - Frame aggregation
  - Channel bonding

802.11ac

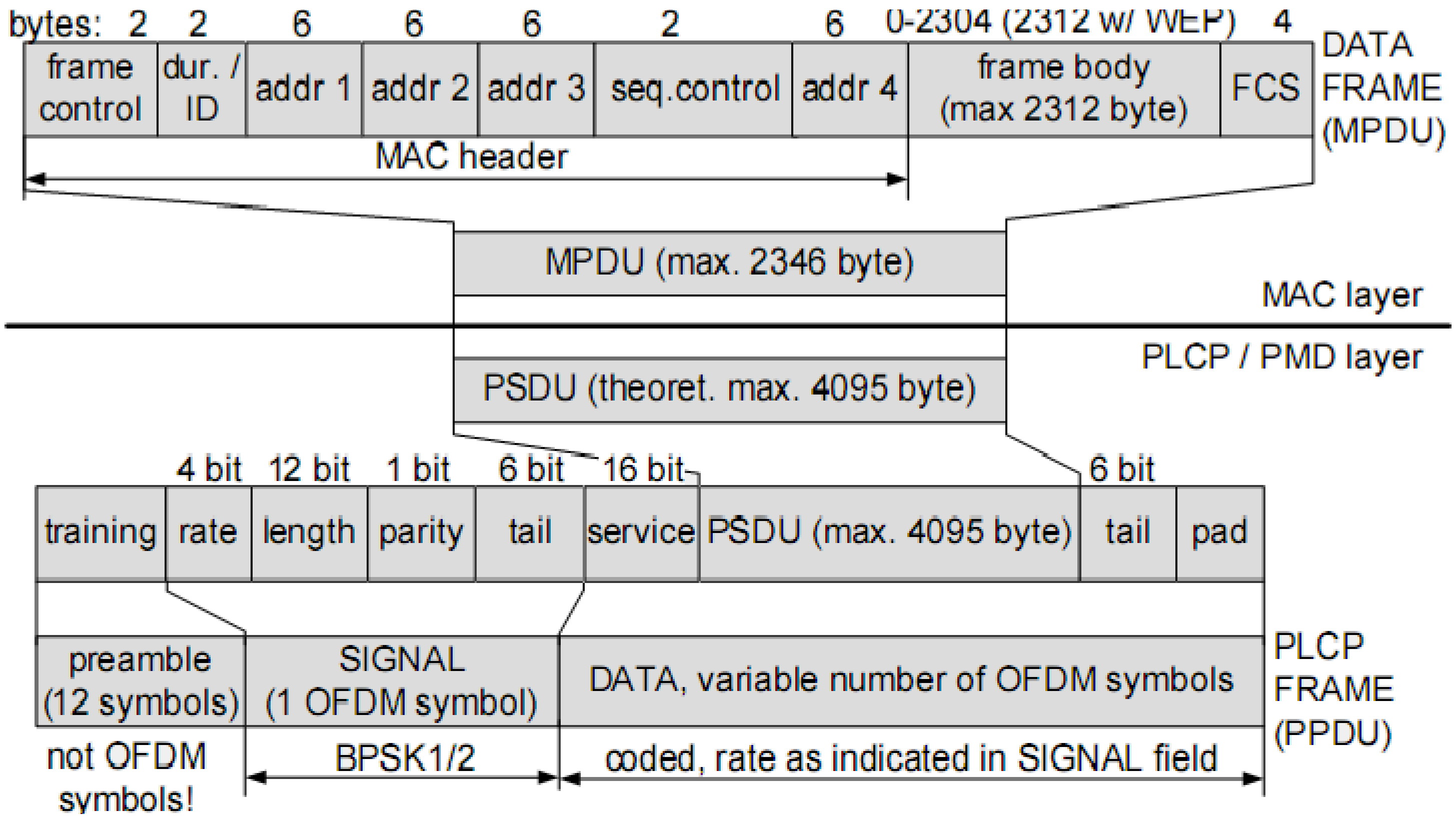
- 5-6 GHz range
- up to 3.4 Gbps
  - MU-MIMO

- all use CSMA/CA for multiple access

802.11 Frame

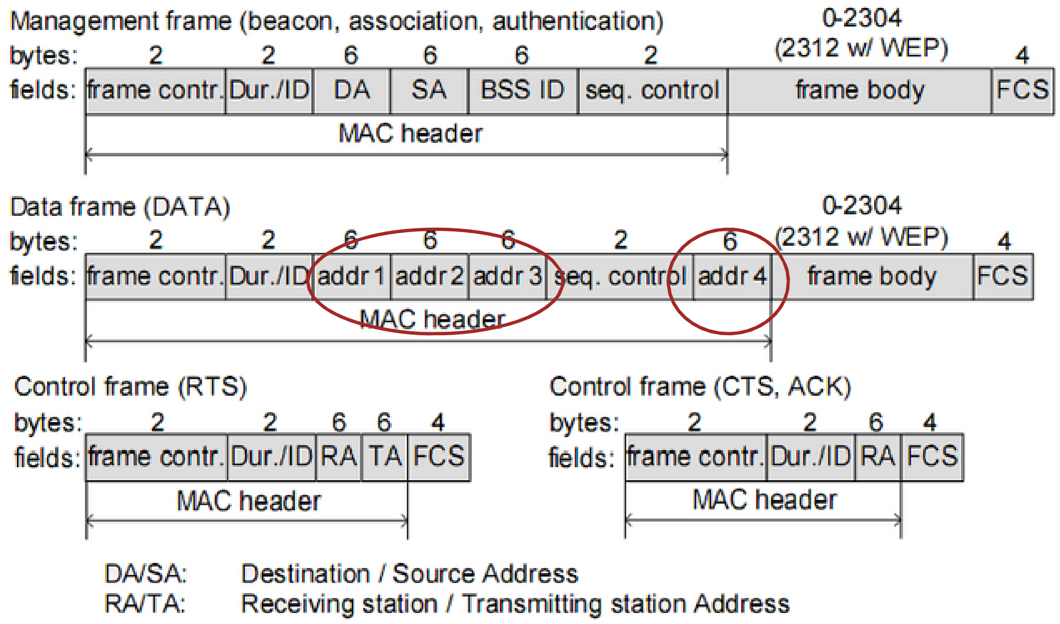
- (in these slides) the OFDM-based 5GHz PHY layer 802.11a is used for description. The alternative PHYs use similar frame formats
- In common, all PHYs consists of 2 sublayers:
  - PLCP
  - PMD
- The PLCP maps the 802.11 MPDUs into PSDUs, a frame suitable for transmitting and receiving data, control and management information between the associated PMD entities.
- The PMD determines the actual transmission scheme, i.e., the way of transmitting and receiving data through the wireless medium.
- The frame that is finally transmitted over the channel is OFDM PLCP frame

802.11 Frame



Frame mapping from MAC down to PLCP

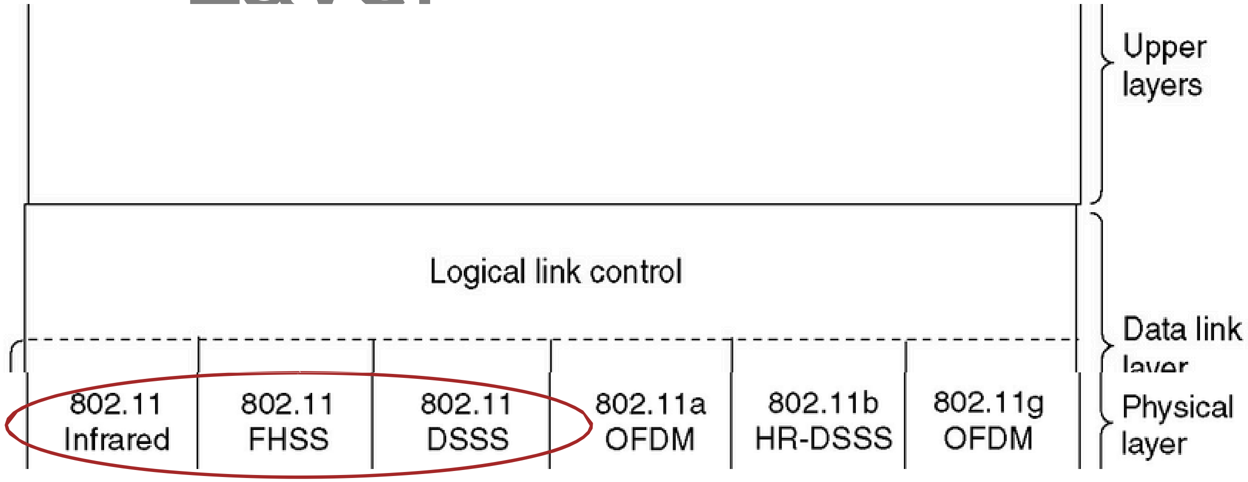
802.11 MAC Frames (MSDU)



- There are 3 types of MPDUs: management, data, and control frames.
- The frame body size of data and management frames are variable.

PHYSICAL LAYER

Physical Layer



- At first, there was 3 different types of PHYs:
  - 2.4 GHz Frequency Hopping Spread Spectrum (FHSS),
  - Direct Sequence Spread Spectrum (DSSS) and
  - Infrared (IR).
- Four more PHYs are defined in supplement standards.

Physical Layer

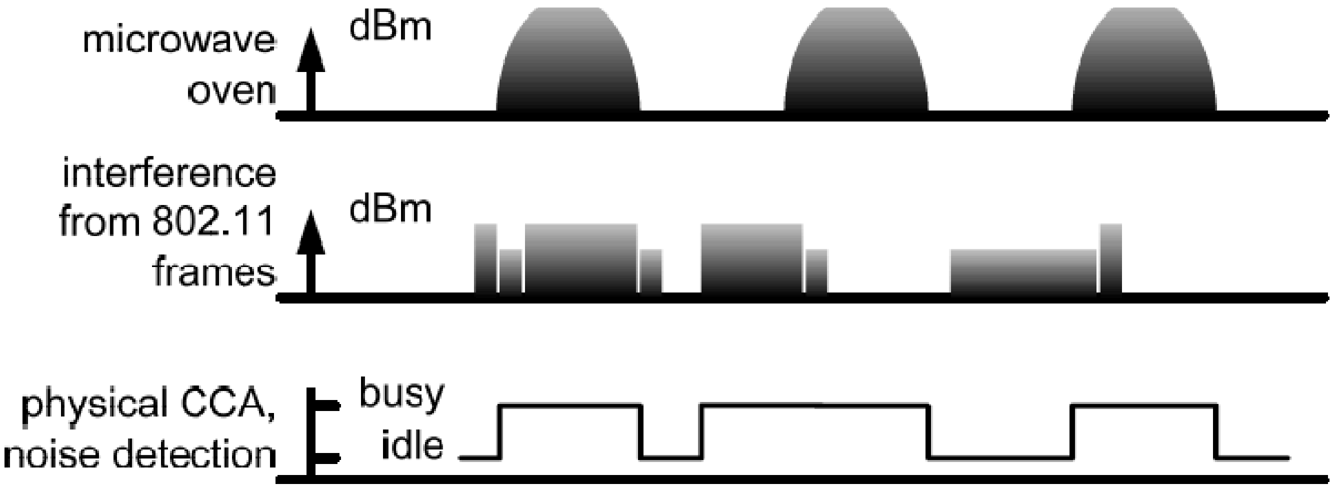
- **FHSS** is a method of transmitting radio signals by rapidly switching a carrier among many frequency channels, using a pseudorandom sequence known to both transmitter and receiver.
- In **DSSS**, all stations operate at the same center frequency. Different spreading codes allow different sets of communicating stations to reduce the mutual interference.
- **IR** physical layer so far has not commercially successful, but may be attractive for future residential in-house applications.

Physical Layer

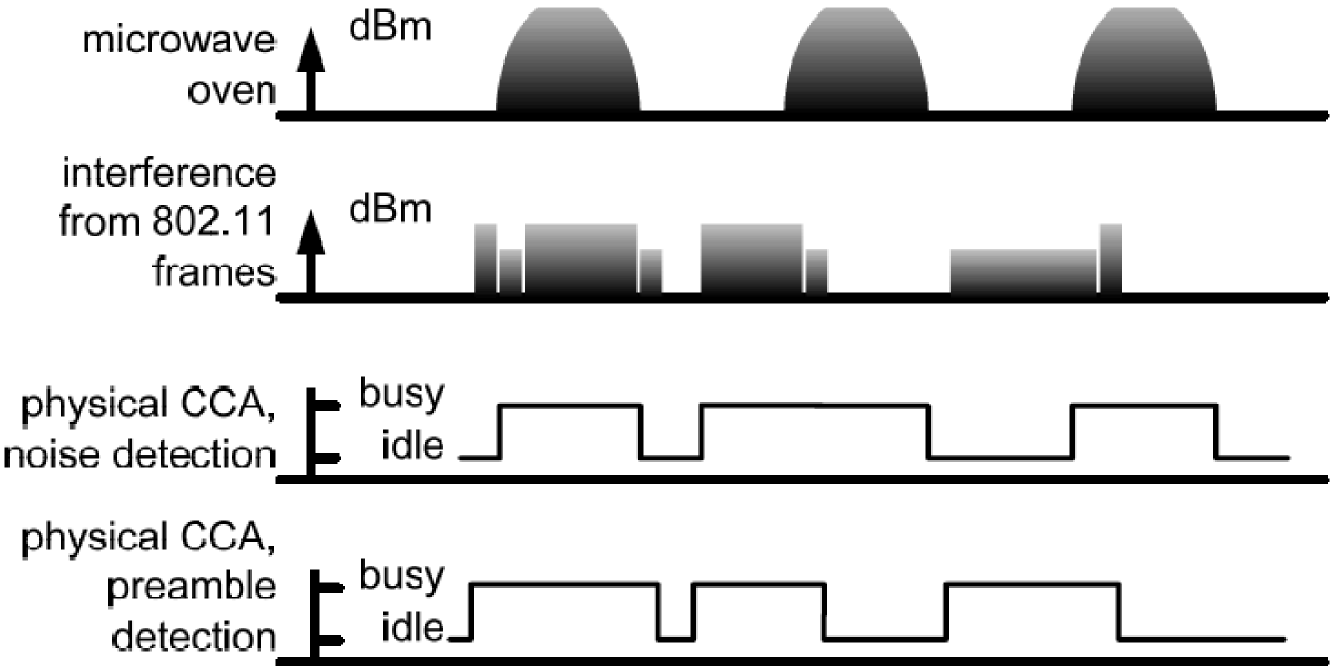
- High Rate Direct Sequence Spread Spectrum (HR/DSSS) transmission mode:
  - uses Complementary Code Keying (CCK). The code set is richer than the set of Walsh codes that are used in DSSS the higher data rate is achieved.
- OFDM is the transmission scheme more and more used by wireless systems.

MEDIUM ACCESS CONTROL PROTOCOL

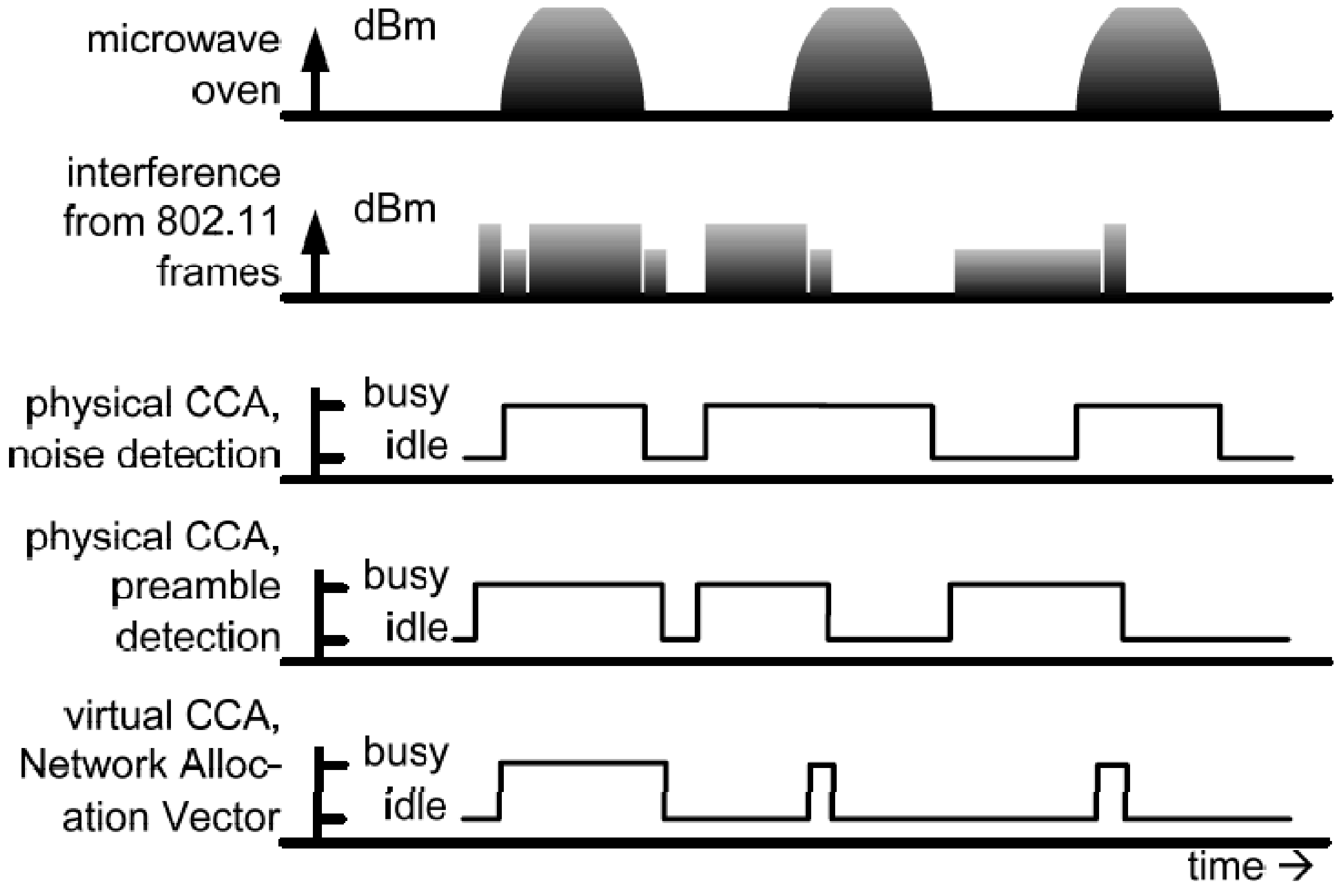
Listen Before



Listen Before



Listen Before



After fully receiving the frame, the node can set NAV

- CCA – Clear Channel Assessment

Listen Before Talk

- The channel sensing function is called **Clear Channel Assessment (CCA).**
  - It uses a power threshold (-82dBm, possibly implementation dependent).
  - If a station detects a signal with power larger than this threshold, the radio channel is assumed to be busy
  - Otherwise the channel is idle.
- There are different variants of CCA (see the previous Fig.)
  - Ordinary noise detection
  - Only signals from other 802.11

Listen Before

- The Network Allocation Vector (NAV) is an addition to the **physical sensing** of the radio channel.
- It is referred to as **virtual carrier sensing** and in fact has the function of reserving the channel for some time.
- It is a timer which decrements irrespectively of the status of the medium.
- As long as the NAV is set or the CCA sensed the radio channel as being busy, a station is not allowed to initiate transmissions.

802.11 Medium Access Control Protocol

- 802.11 MAC protocol is built with the help of 2 coordination functions:
  1. DCF for traffic without QoS (asynchronous services) and
  2. PCF for traffic with QoS requirements (synchronous services).
- In the following,
  - We discuss Distributed Coordination Function (DCF)
  - an infrastructure-based BSS
- The basic 802.11 MAC protocol is the DCF that works as a **listen-before-talk** scheme, based on the Carrier Sense Multiple Access (CSMA)

IEEE 802.11
Distributed Coordination Function (DCF)

Distributed Coordination Function (DCF)

- 802.11 DCF : based on the CSMA - sense before transmitting
  - *No* collision detection (CD)!
    - Difficult to receive (sense collisions) when transmitting due to weak received signals (fading)
    - Can't sense all collisions in any case: hidden terminal, fading
    - Goal: *avoid collisions:* CSMA/C(ollision)A(voidance)
  - Link layer ACKnowledgments/Retransmissions (ARQ)
    - High bit error rates

C
B
A

A
B
C

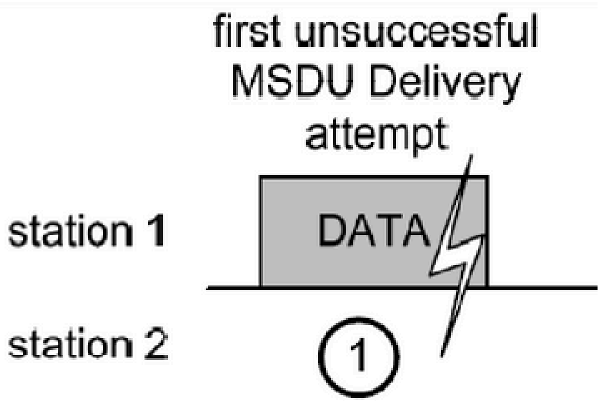
A's signal strength

C's signal strength

space

Timing and Interframe Spaces

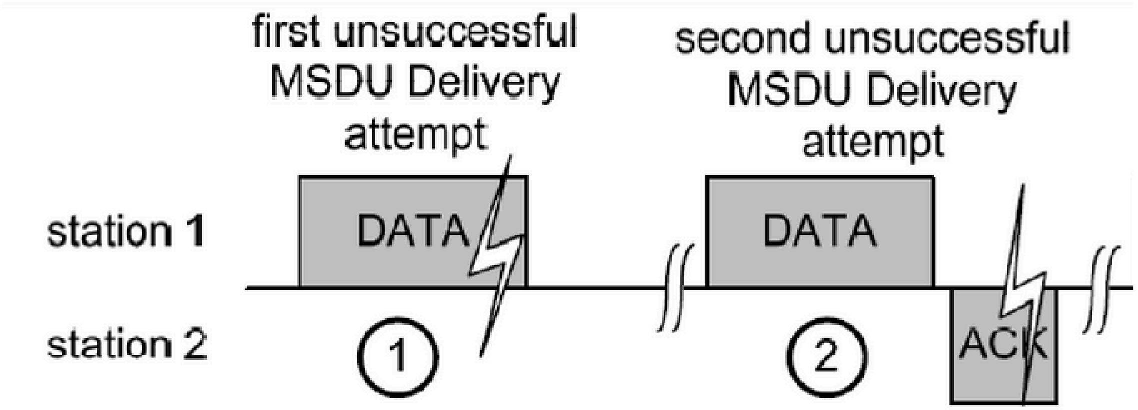
Frame Exchange Scenarios



- First attempt: DATA frame from station 1 is not received, no ACK

Timing and Interframe Spaces

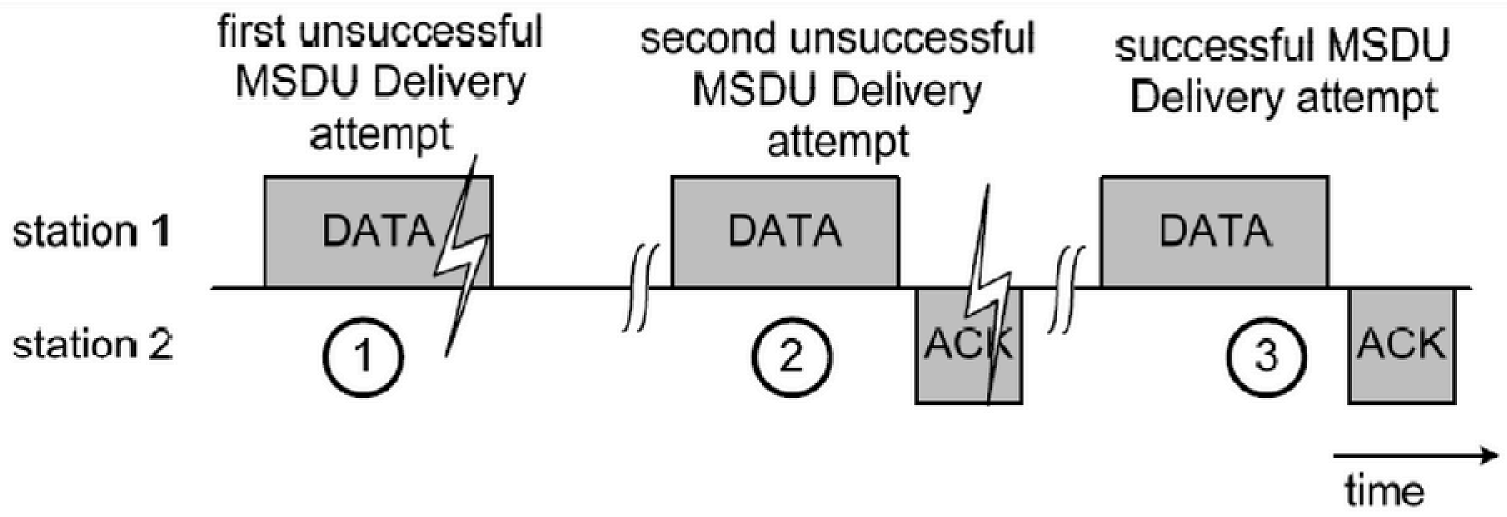
Frame Exchange Scenarios



- First attempt: DATA frame from station 1 is not received, no ACK
- Second attempt: because of the missing ACK from station 2, the station 1 retransmits the DATA frame. This is successfully received, the corresponding ACK is sent as response, but missed by station 1 (for example because of interference)

Timing and Interframe Spaces

Frame Exchange Scenarios



- First attempt: DATA frame from station 1 is not received, no ACK
- Second attempt: because of the missing ACK from station 2, the station 1 retransmits the DATA frame. This is successfully received, the corresponding ACK is sent as response, but missed by station 1 (for example because of interference)
- Third and successful attempt: DATA and ACK are both successfully received

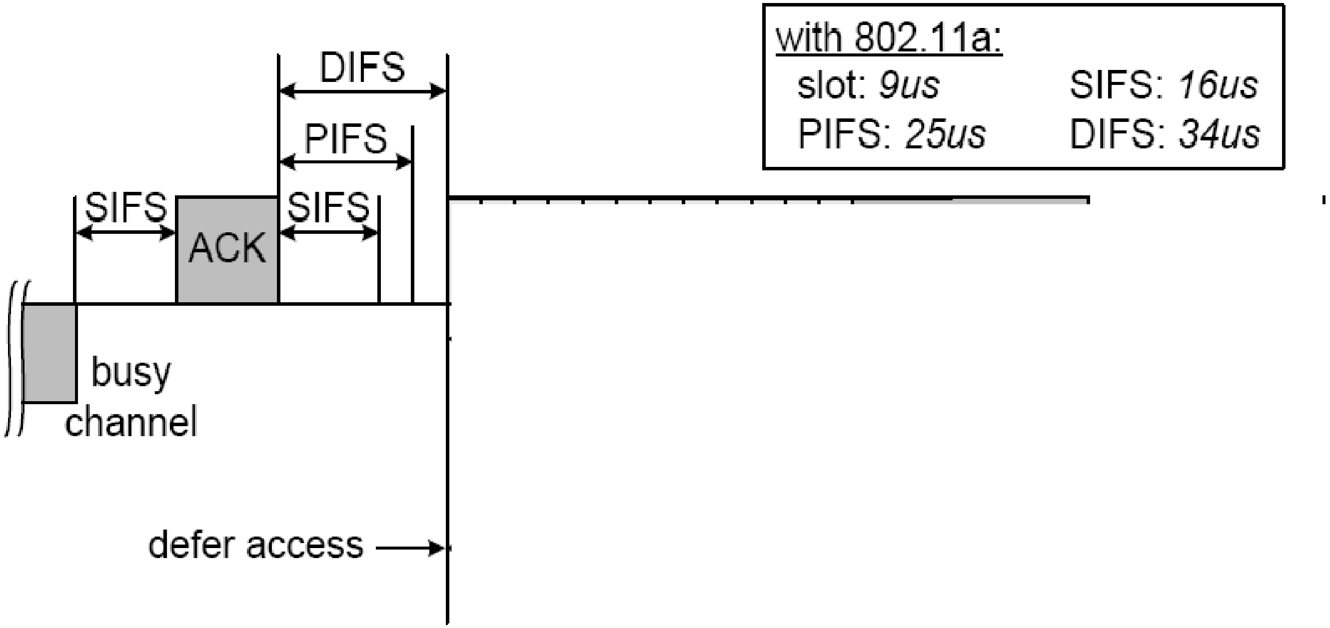
Timing and Interframe Spaces

- The time between 2 MAC frames is called the **Interframe Space (IFS).**
- 802.11 defines 4 different IFSs:
  - Short Interframe Space (SIFS)
  - Point Coordination Function Interframe Space (PIFS)
  - Distributed Coordination Function Interframe Space (DIFS)
  - Extended Interframe Spce (EIFS)
- These IFSs represent three different levels for medium access (See the Fig. in the following)
- The shorter the IFS, the higher priority in medium access.
- All interframe spaces are independent of the channel data rate.

Timing and Interframe Spaces

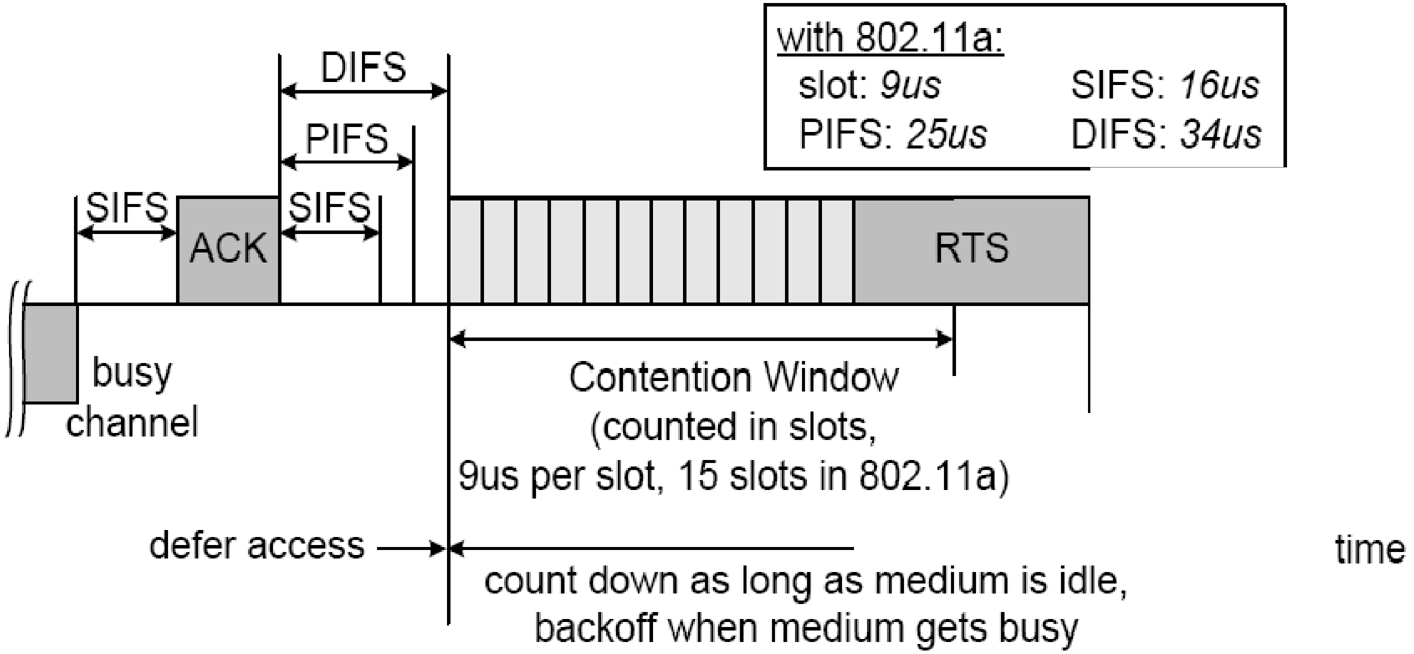
- aSlotTime
  - is used to calculate the IFSs.
  - SIFS and aSlotTime are the basis of all other duration.
  - In 802.11a, aSlotTime=9μs
- SIFS
  - is used to prioritize the immediate ACK of a data frame,
  - the response (CTS) to a RTS,
  - a subsequent MPDU of a fragmented MSDU,
  - response to any polling using the PCF, and
  - any frames of the AP during the CFP.
  - SIFS = 16μs for 802.11a
- PIFS
  - used by stations operating under the PCF to obtain channel access with highest priority.
  - PIFS = SIFS + aSlotTime.
  - PIFS = 25μs for 802.11a
- DIFS
  - used by stations operating under the DCF to obtain channel access.
  - DIFS = SIFS + 2 * aSlotTime
  - DIFS = 34μs for 802.11a

Timing and Interframe Spaces



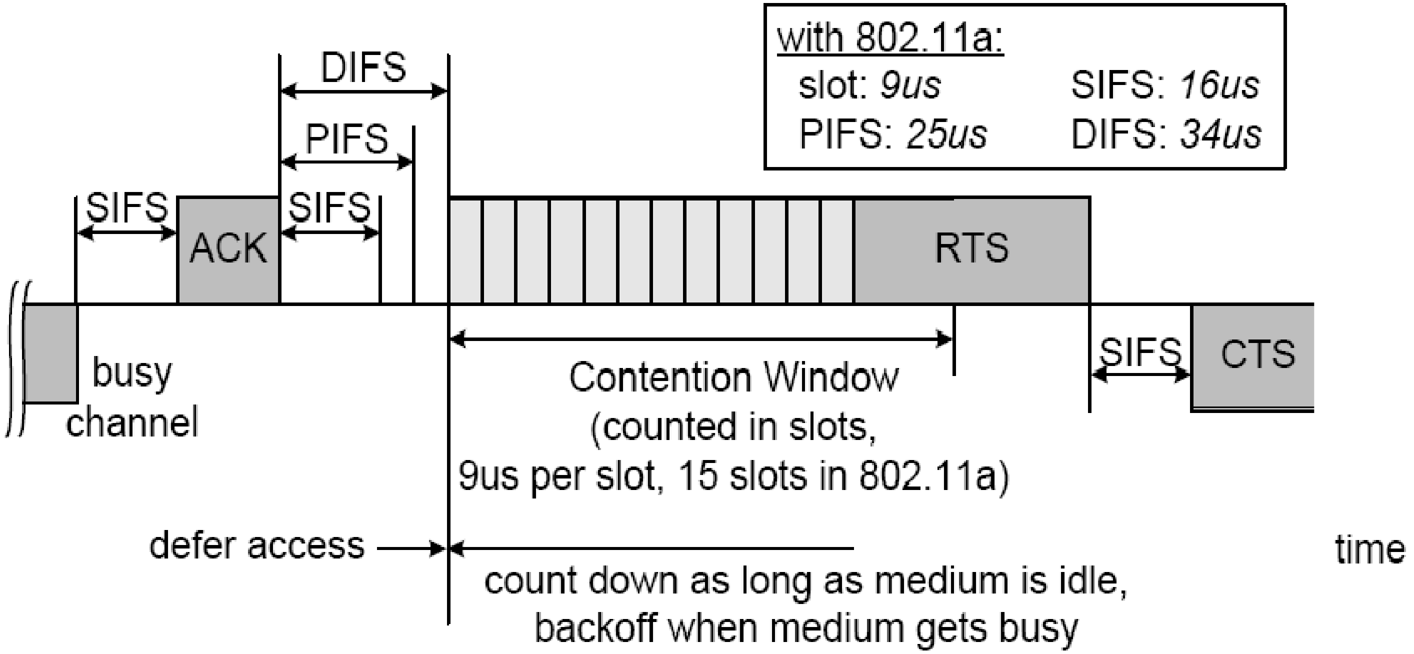
- SIFS = Short Interframe Space (no CCA carrier sensing)
- PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
- DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is too short for CCA, hence only needed for transceiver turnaround

Timing and Interframe Spaces



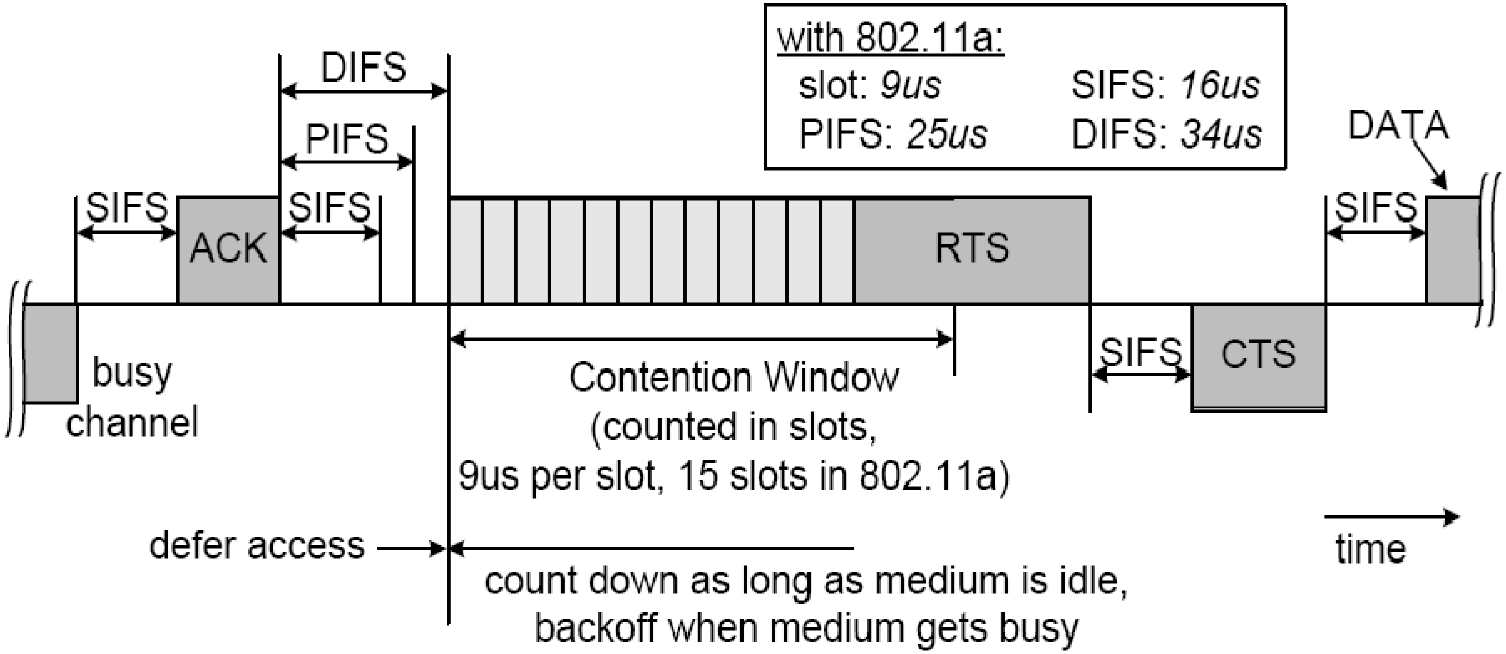
- SIFS = Short Interframe Space (no CCA carrier sensing)
- PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
- DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is too short for CCA, hence only needed for transceiver turnaround

Timing and Interframe Spaces



- SIFS = Short Interframe Space (no CCA carrier sensing)
- PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
- DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is too short for CCA, hence only needed for transceiver turnaround

Timing and Interframe Spaces



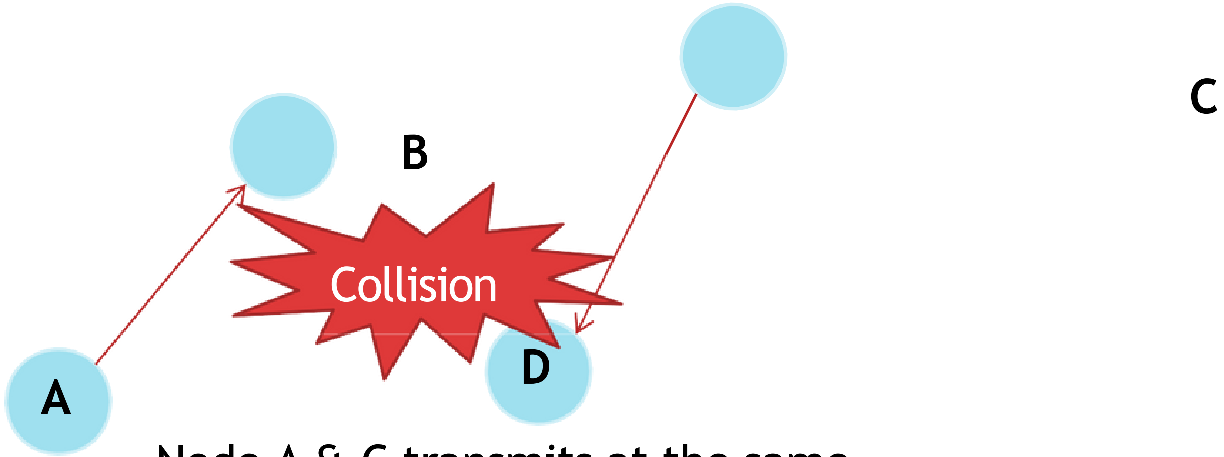
- SIFS = Short Interframe Space (no CCA carrier sensing)
- PIFS = PCF Interframe Space = SIFS + aSlotTime (with CCA)
- DIFS = DCF Interframe Space = PIFS + aSlotTime (with CCA)
- SIFS is too short for CCA, hence only needed for transceiver turnaround

Timing and Interframe Spaces

- EIFS (Extended IFS): used instead of DIFS when multiple stations initiated frame exchanges at different starting times, typically due to they are **hidden** from each other.
- The EIFS results in a longer deferral from channel access, which gives other stations a higher priority in medium access.
- As soon as 1 other frame is received correctly, DIFS is used again.
- EIFS is around 200μs for 802.11a

Collision

- A **collision** happens when there are more than 1 station attempts to transmit at the same time.
- In wireless communication, transmitter cannot detect a collision at receiver while transmitting. To account for this,
  802.11 is based on CSMA/CA.



Node A & C transmits at the same

Collision Avoidance

- If 2 or more stations detect the channel as being idle at the same time, inevitably a collision occurs when these stations initiate transmission at the same time.
- 802.11 defines a CA mechanism to reduce the probability of such collisions.

Collision

- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- A station having MSDU to deliver has to keep sensing the channel for an additional random time (=backoff) after detecting the channel as being idle for at least a DIFS (=34μs).
- Duration of Backoff =

  Random Backoff count,r x aSlotTime. (r=integer)
- Each station maintains a so-called **Contention Window (CW)**, which is used to determine n.
  - $CW = min(2^{(n-1)} * CWmin - 1, \; CWmax)$
  - Random Backoff count,r = Rand(0 ~ CW)
- n= number of attempt
- In IEEE 802.11a *CWmin =16 ,* CWmax = 1024

Avoiding collisions (more)

idea: allow sender to "reserve" channel rather than random access of data frames: avoid collisions of long data frames

sender first transmits *small* request-to-send (RTS) packets to BS using CSMA

RTSs may still collide with each other (but they're short)

BS broadcasts clear-to-send CTS in response to RTS

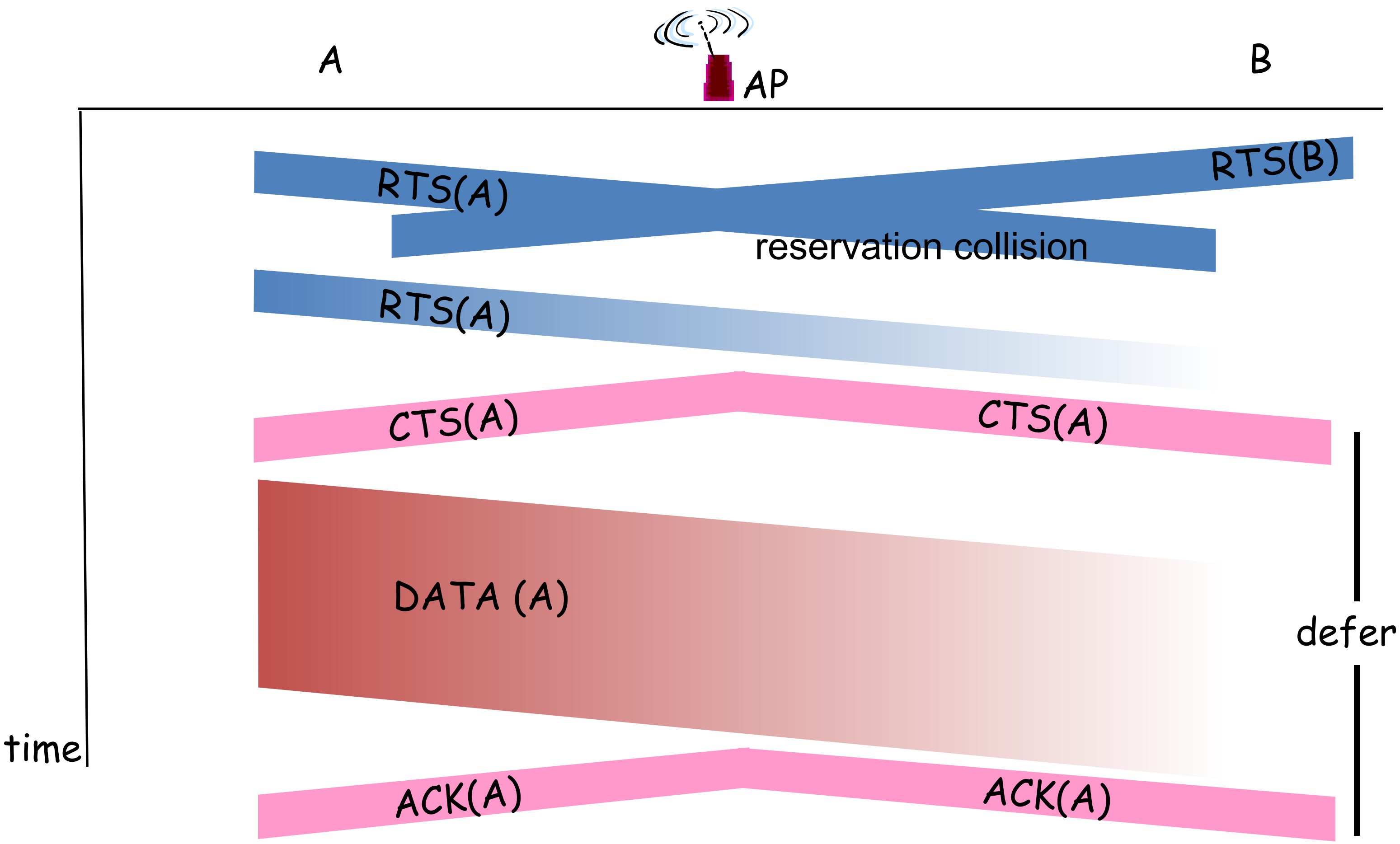
CTS heard by all nodes

sender transmits data frame

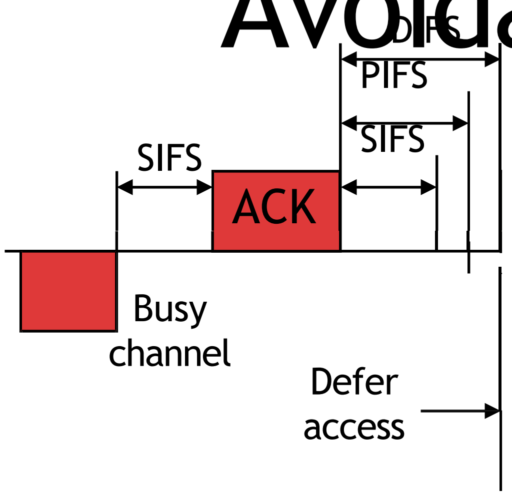
other stations defer transmissions

avoid data frame collisions completely using small reservation packets!

Collision Avoidance: RTS-CTS exchange

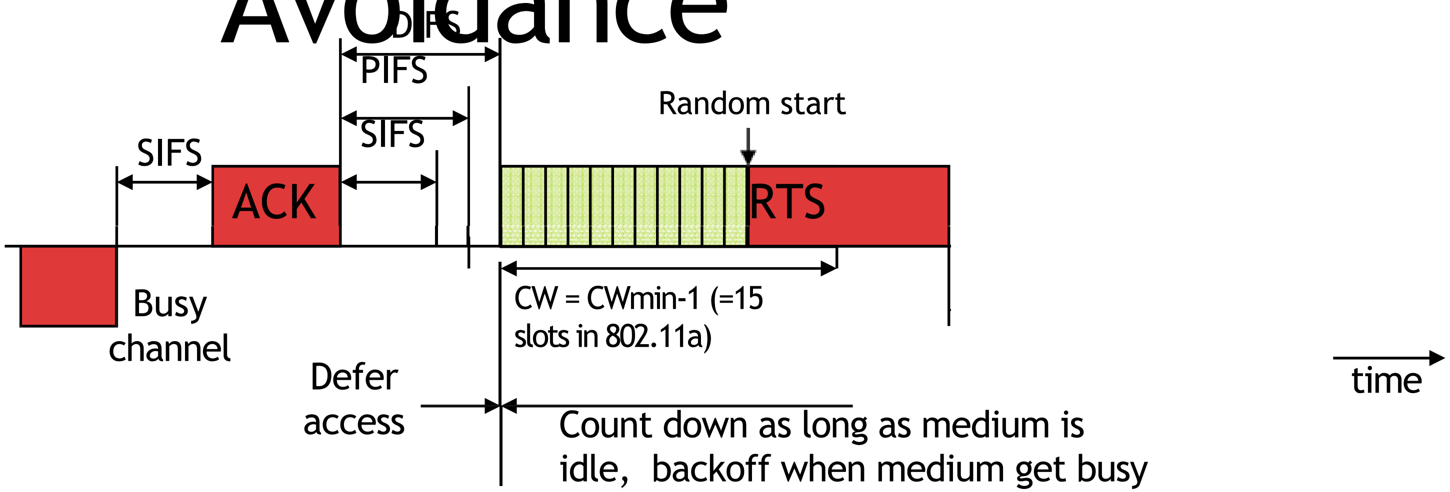


Collision Avoidance



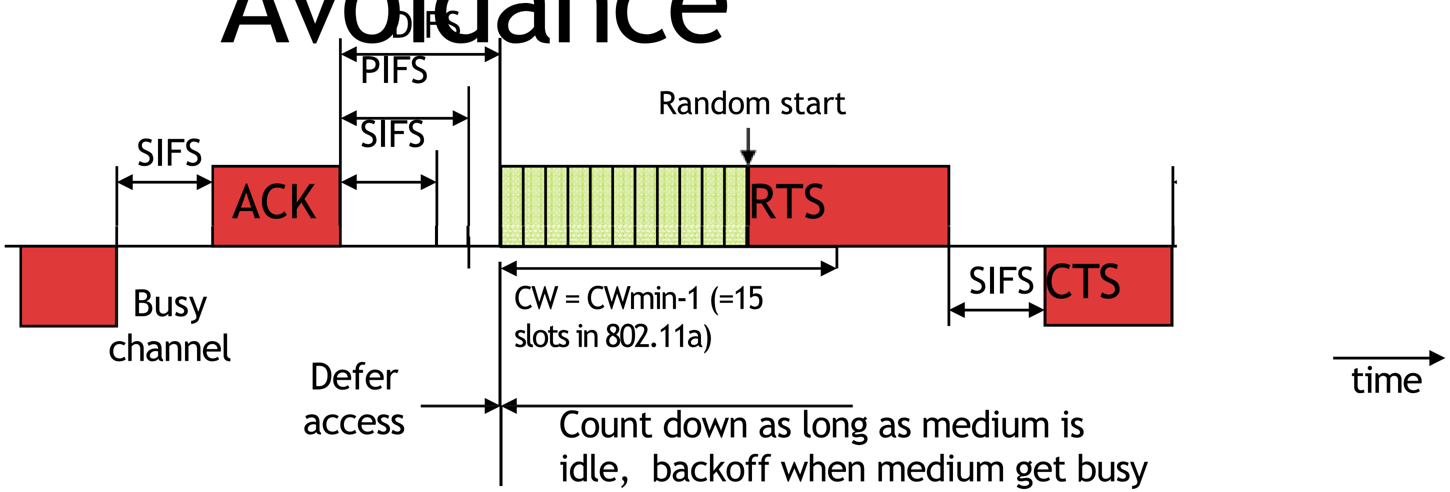
- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- Ex: After a successful frame exchange, a station wants to send another data frame. It starts transmitting RTS after sensing the channel is idle for a duration equal to DIFS and its following backoff slots.

Collision Avoidance



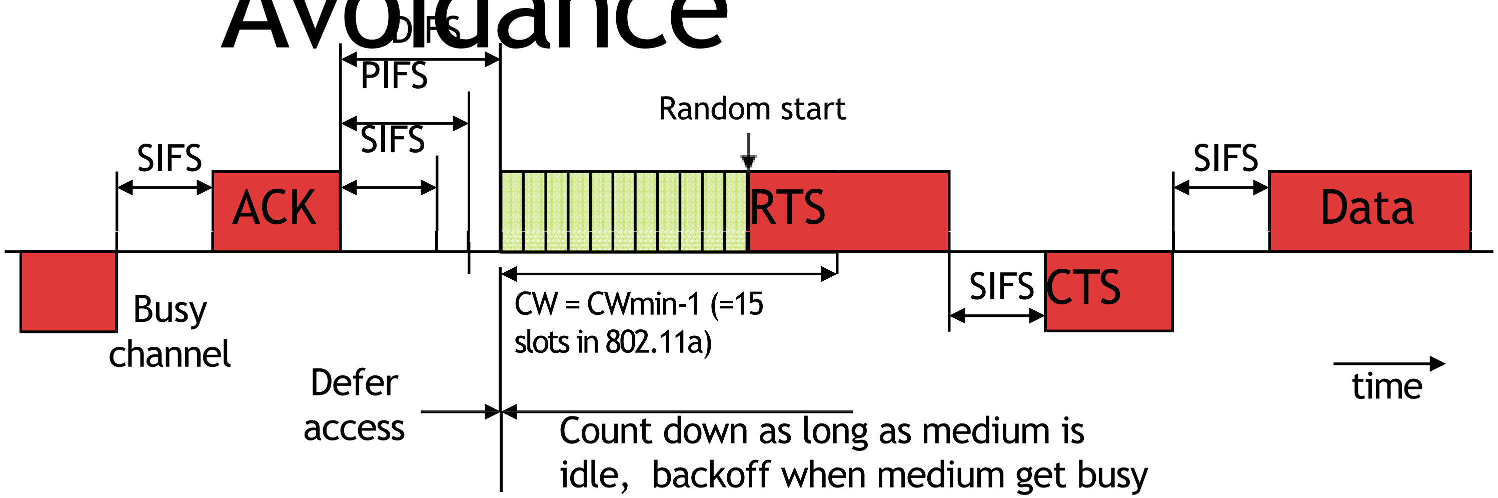
- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- Ex: After a successful frame exchange, a station wants to send another data frame. It starts transmitting RTS after sensing the channel is idle for a duration equal to DIFS and its following backoff slots.

Collision Avoidance



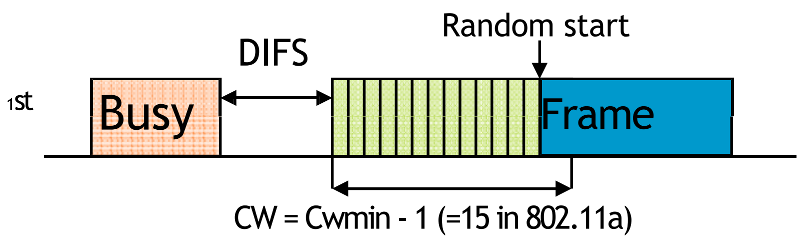
- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- Ex: After a successful frame exchange, a station wants to send another data frame. It starts transmitting RTS after sensing the channel is idle for a duration equal to DIFS and its following backoff slots.

Collision Avoidance

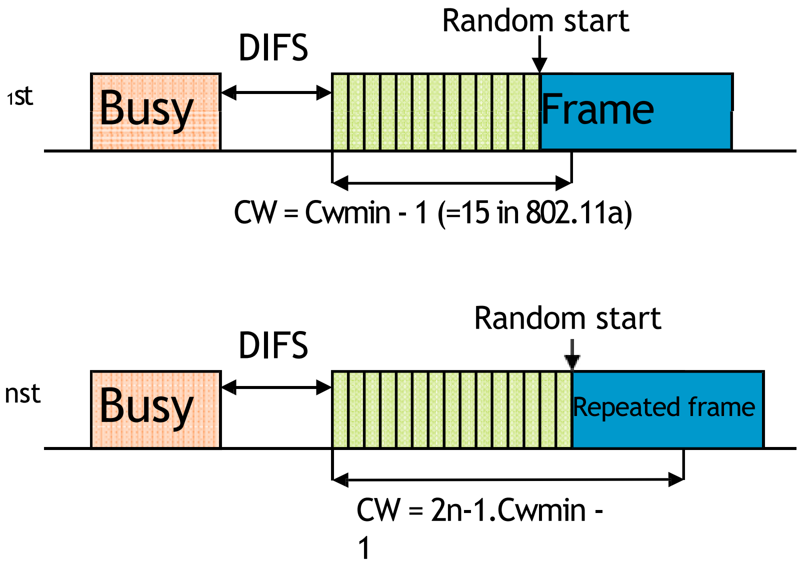


- As part of CA, a station performs the **backoff procedure** before starting a transmission.
- Ex: After a successful frame exchange, a station wants to send another data frame. It starts transmitting RTS after sensing the channel is idle for a duration equal to DIFS and its following backoff slots.

Collision Avoidance



Collision Avoidance

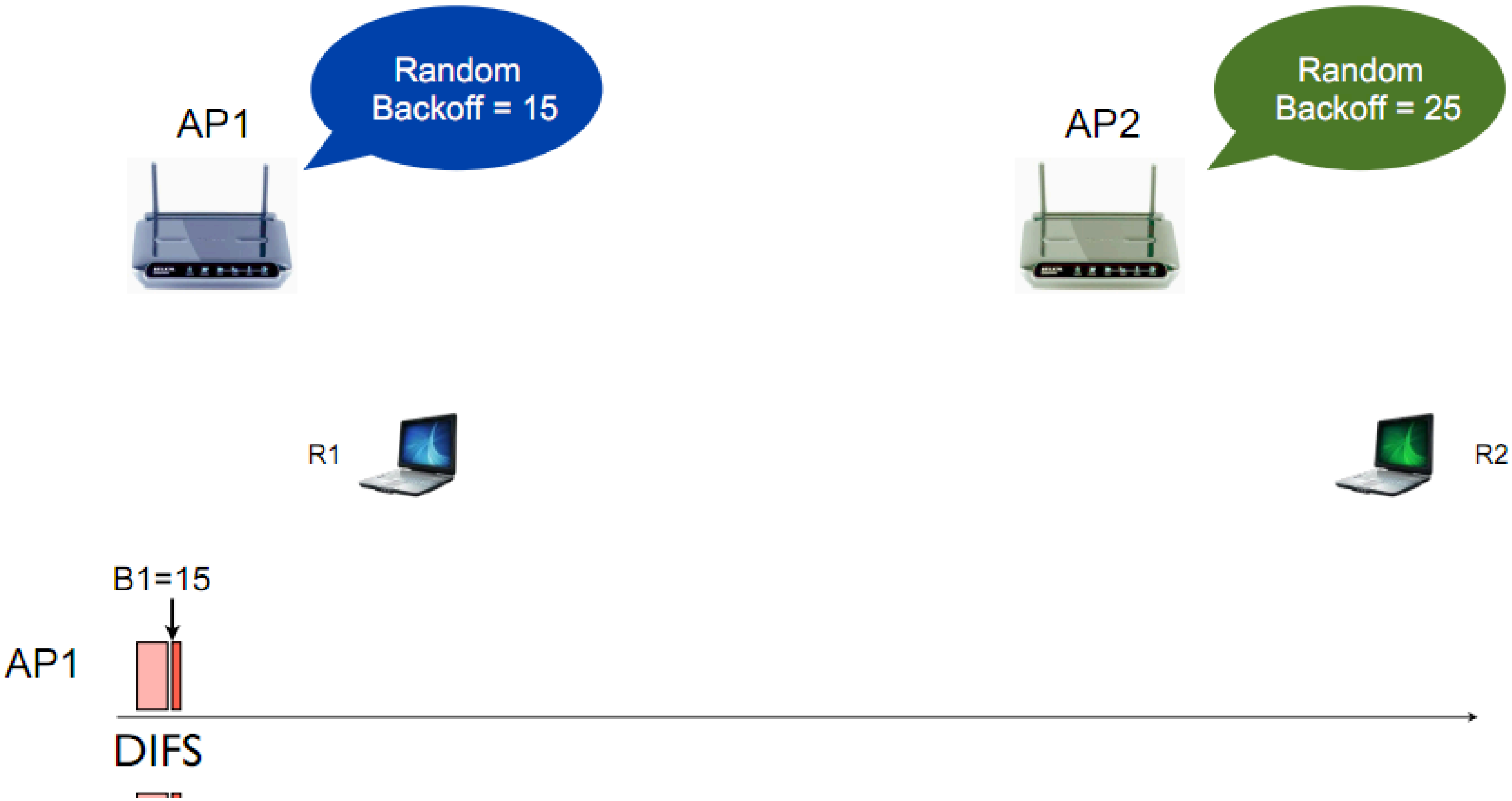


- After an unsuccessful transmission, the next backoff is performed with a doubled size of the contention window. This reduces the collision probability if there are multiple stations attempting to access the channel.

Collision Avoidance

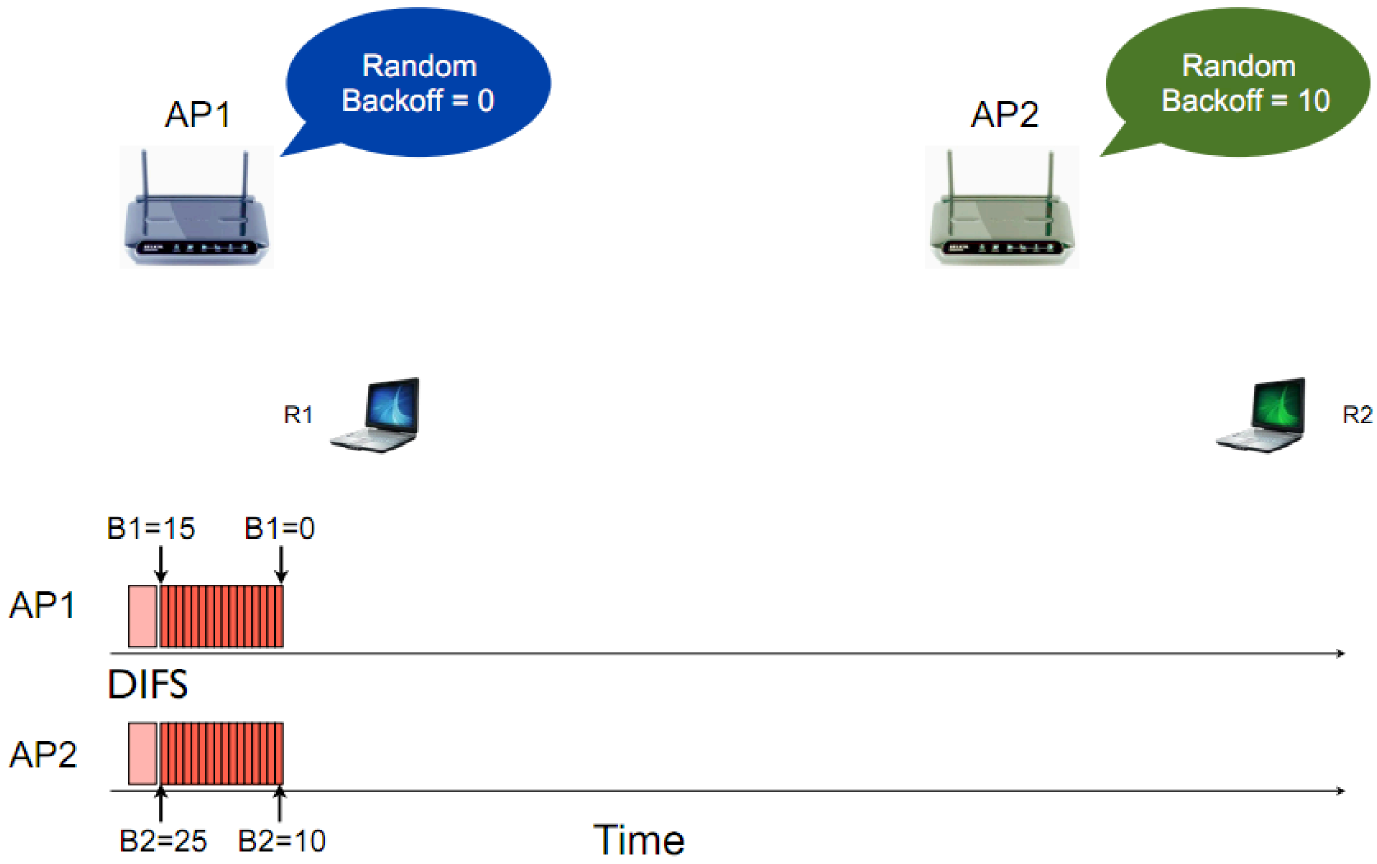
- The station that deferred from channel access during the channel busy period do not select a new random backoff time, but continue to count down the time of previous backoff after sensing the channel as idle again.
- In this way, stations, which deferred from medium access because their random backoff was larger than other's, are given a higher priority when they resume the transmission attempt.

802.11 Backoff Example

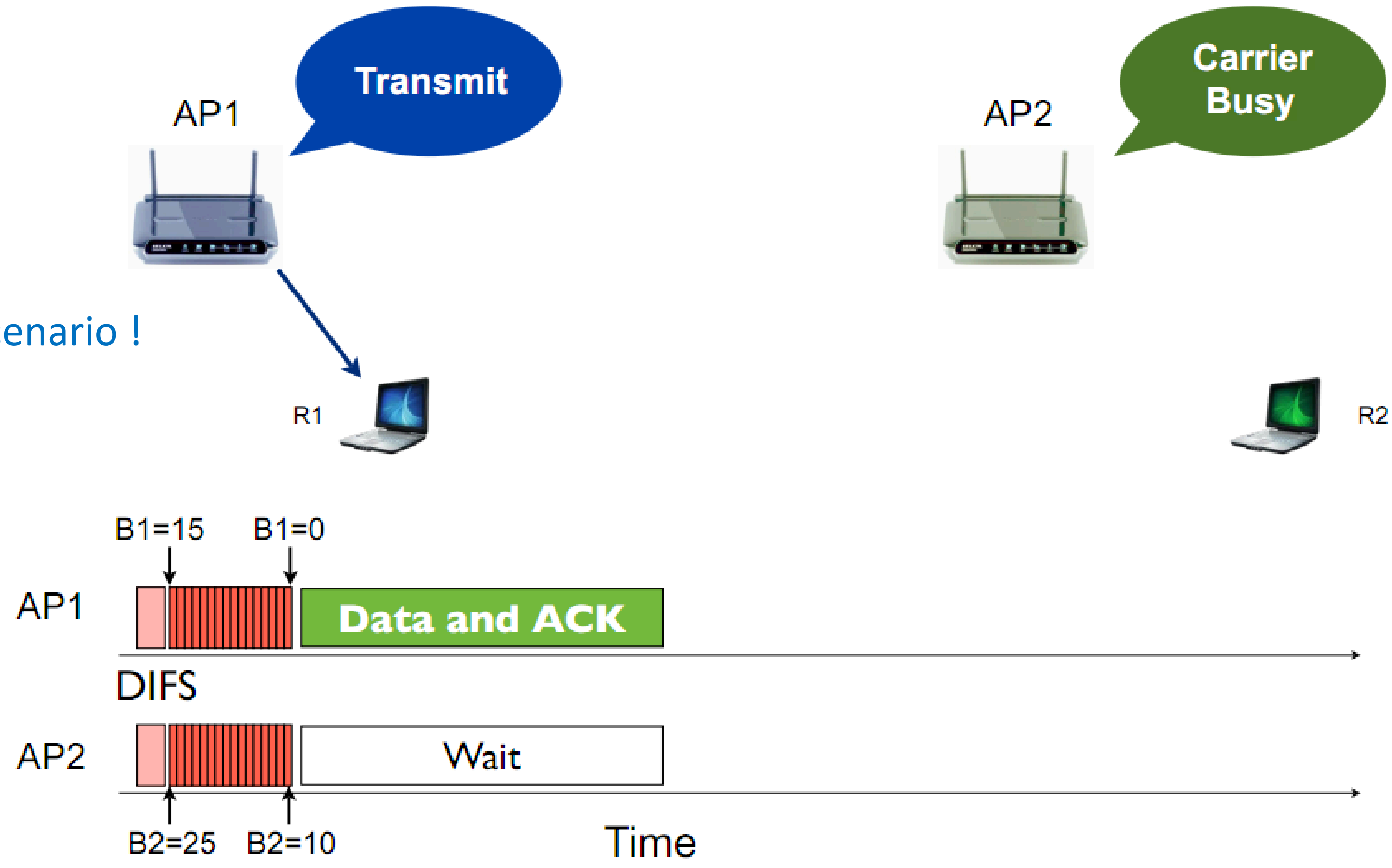


802.11 Backoff Example

After 15 slot countdown



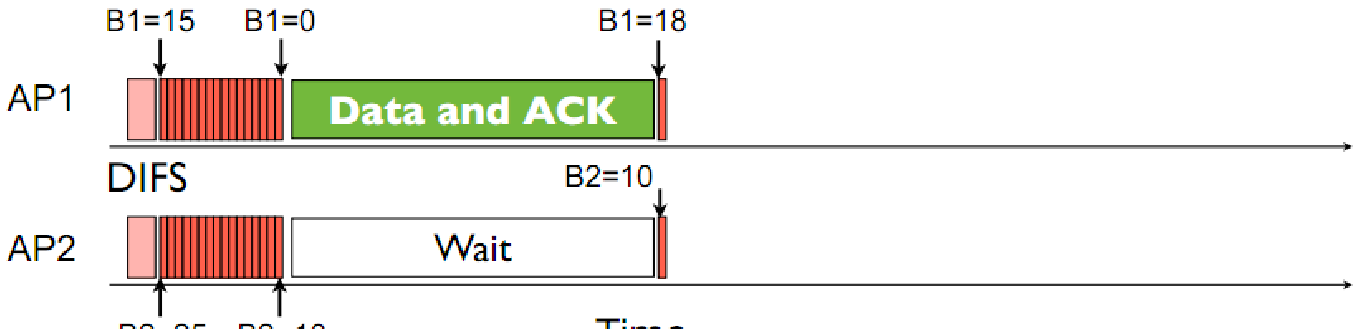
802.11 Backoff Example



RTS, CTS Disabled Scenario !

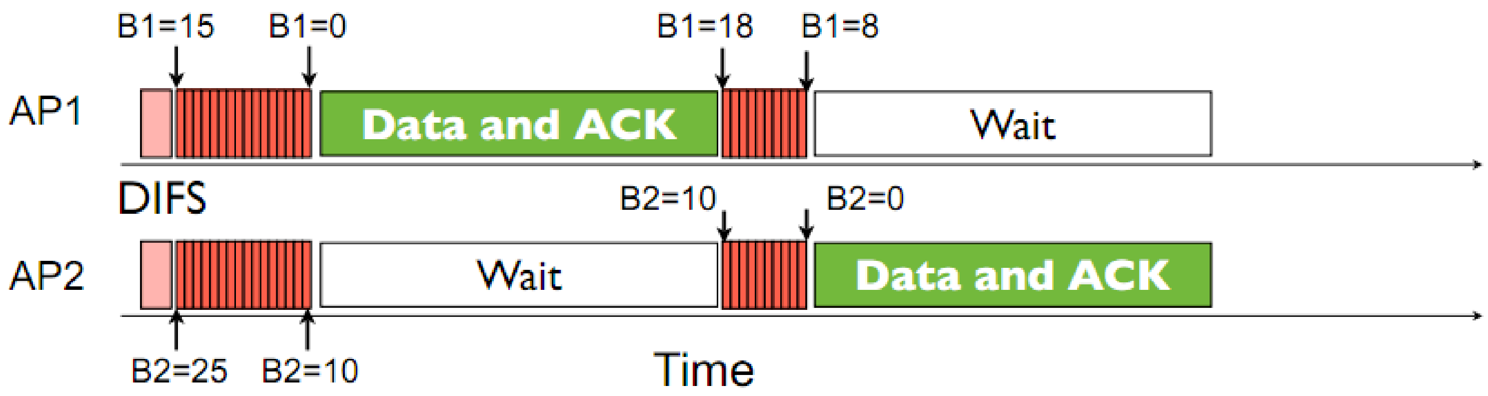
802.11 Backoff Example

RTS, CTS Disabled Scenario !



802.11 Backoff Example

RTS, CTS Disabled Scenario !



Distributed Coordination Function (DCF)

802.11 sender

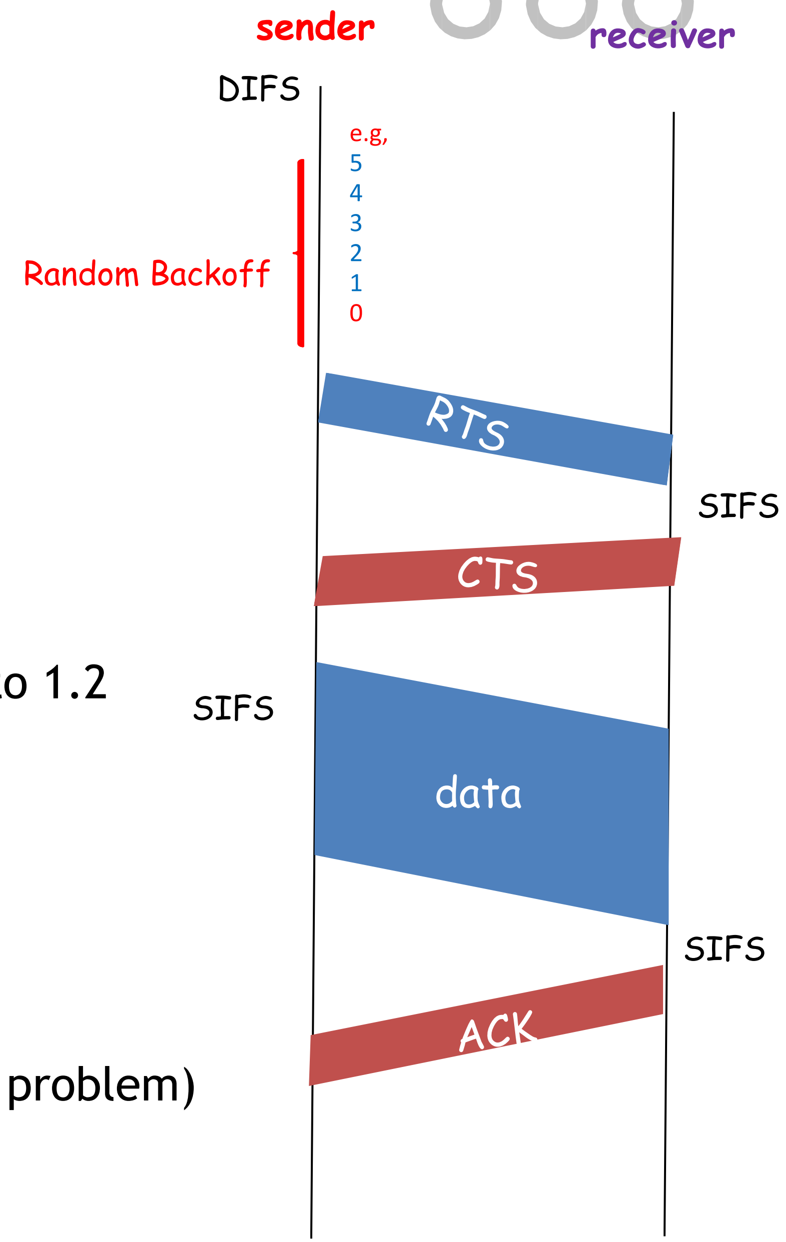
1.1 if sense channel busy then freezes

1.2 if sense channel idle for **DIFS** then

2.1 start random backoff time

- timer counts down while channel idle,
- freezes when busy,
- resumes if idle for DIFS
- transmit RTS when timer expires (why?)

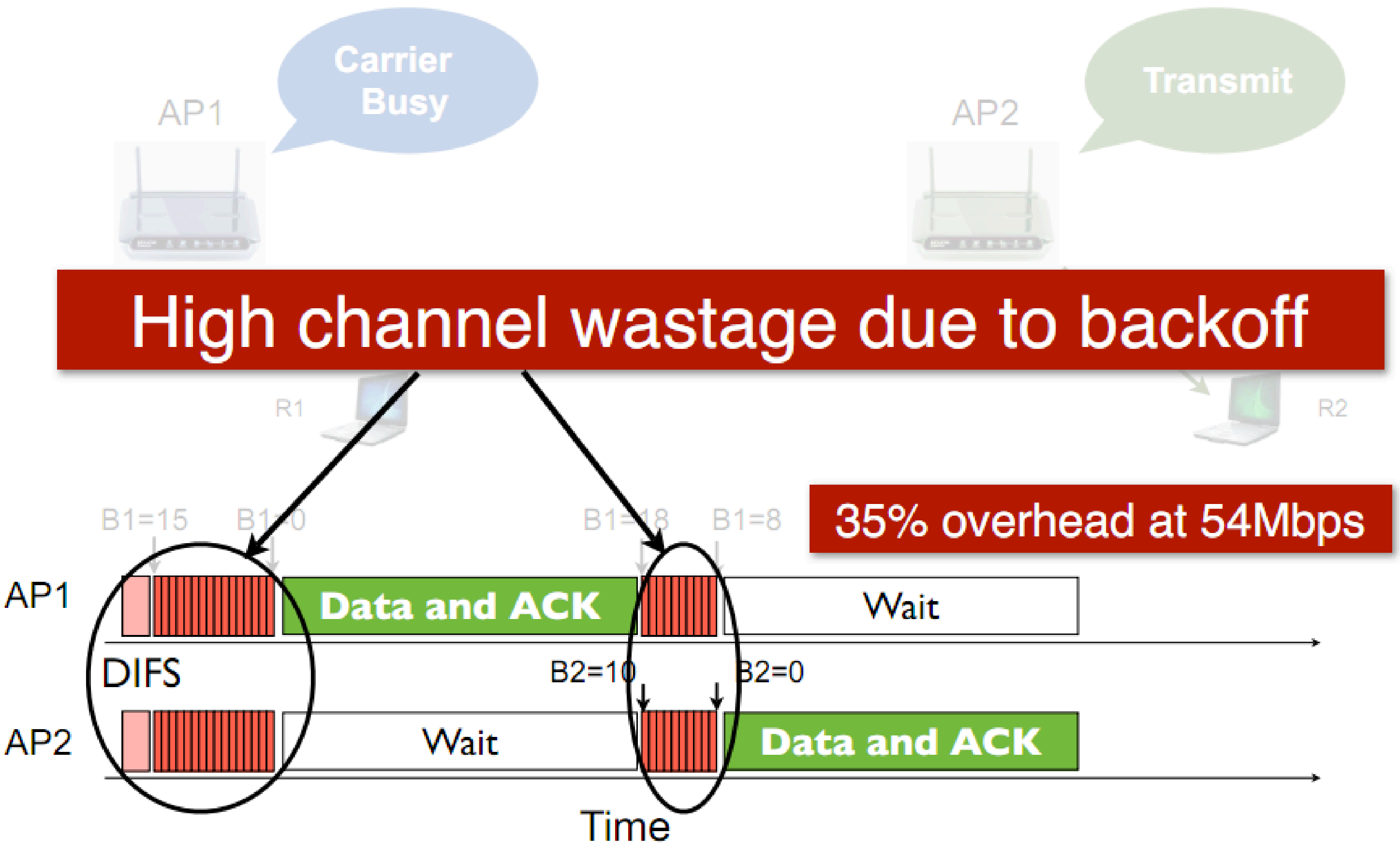
3. if no CTS, then go to 1.2 (why?)
4. if CTS, then wait SIFS, and transmit DATA
5. if no ACK, increase random backoff interval, go to 1.2 (why?)



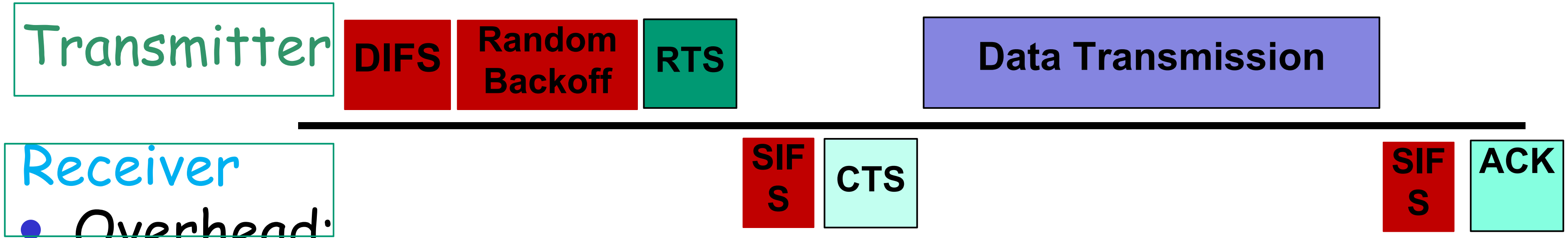
802.11 receiver

- if RTS frame received OK
- return CTS after **SIFS**
- if Data frame received OK
- return ACK after **SIFS** (ACK needed due to hidden terminal problem)

802.11 Medium Access Overhead



802.11 Medium Access Overhead



- Overhead:
  - DIFS
  - Random backoff
  - ACK/SIFS
  - Optional RTS/CTS handshake before transmission of data packet (often disabled due to its overhead)
  - Header overhead
- 802.11 has room for improvement. How?
  - (Will be discussed in coming lectures)

802.11 Medium Access Overhead

802.11 Medium Access Overhead

Significant Inefficiencies of 802.11 DCF :

- **Idle Channel Time:** During the countdown, the channel remains idle, even though no transmission occurs. This results in underutilization of available bandwidth.
- **Collision Probability**: When two or more nodes pick the same backoff value, simultaneous transmission leads to **packet collisions**. These collisions cause retransmissions and an **exponential increase in CW**, further delaying future transmissions.

- **Exponential Backoff Under Congestion:** In IEEE 802.11, Binary Exponential Backoff (BEB) increases CW size as:
  - *CW = min(2^n * CWmin – 1,* Cwmax)
- for each failed attempt (up to CWmax = 1024).
- At 10 active nodes, probability of at least two nodes choosing the same slot exceeds 40%, drastically reducing throughput and increasing latency.

802.11 Medium Access Overhead

Check for More Details: Lecture-6.2 Inefficiency of Random Access based Multiple Access Protocol.pdf

Recovery Procedure and Retransmissions

- When a frame exchange is not successful, the size of the frame is compared against a threshold value before retransmission
- If Size < threshold, or the frame was RTS
  - increment SRC (Short Retry Counter);
  - else increment LRC (Long Retry Counter).
- The frame is discarded when SRC or LRC reaches a limit (default: SRC limit = 7, LRC limit = 4)
- Whenever an MSDU is successfully transmitted, SRC and LRC are reset.
- The threshold is implementation dependent

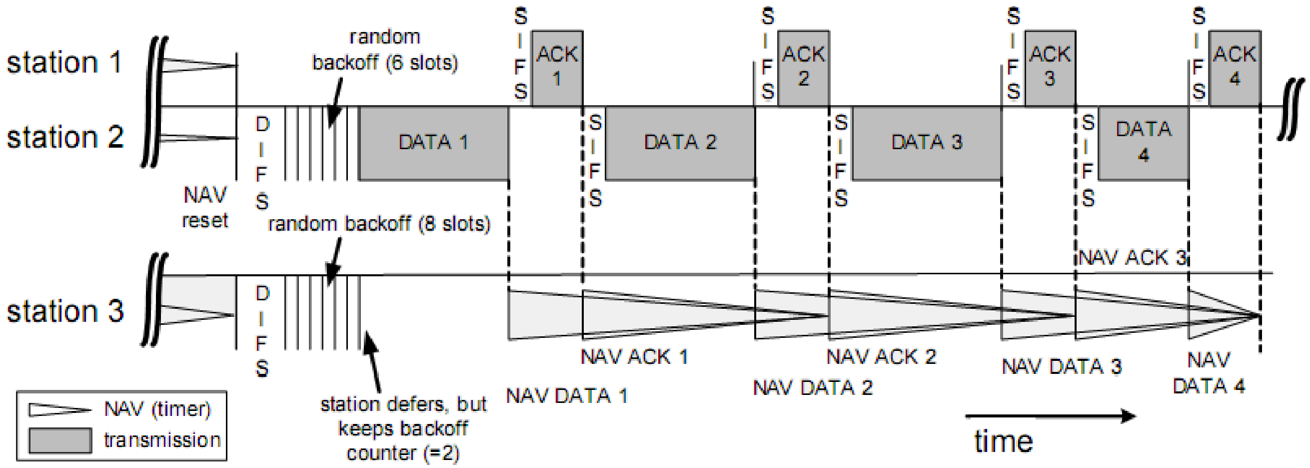
Post-backoff

- After each successful transmission, it is mandatory that another random backoff is performed even if there is no other MSDU to be delivered at the station. This is referred to as **post-backoff** because it is performed after, not before, a transmission.
- It is guaranteed that any frame will be delivered with backoff.
- An MSDU arriving at the station from the higher layer may be transmitted immediately without waiting any time, if the transmission queue is empty, the latest post-backoff has been finished, and at the same time, the channel has been idle for a minimum duration of DIFS.
- This helps to reduce the delivery delay in lightly loaded systems.

Fragmentation

- The process of partitioning an MSDU into smaller MPDUs when its length exceeds a **fragmentation threshold**.
- The benefit of fragmentation is that in the case of failed transmission, the error is detected earlier and there is less data to retransmit.
- It also increases the probability of successful transmission of the MSDU in scenarios where the radio channel characteristics cause higher error probabilities for longer frames than for shorter ones.
- The obvious drawback is the increased overhead.

Fragmentation



- The process of recombining MPDUs into a single MSDU is called defragmentation.
- Only MPDUs with a unicast receiver address may be fragmented.
- Note: The maximum length of an MSDU is limited to 2346 byte.

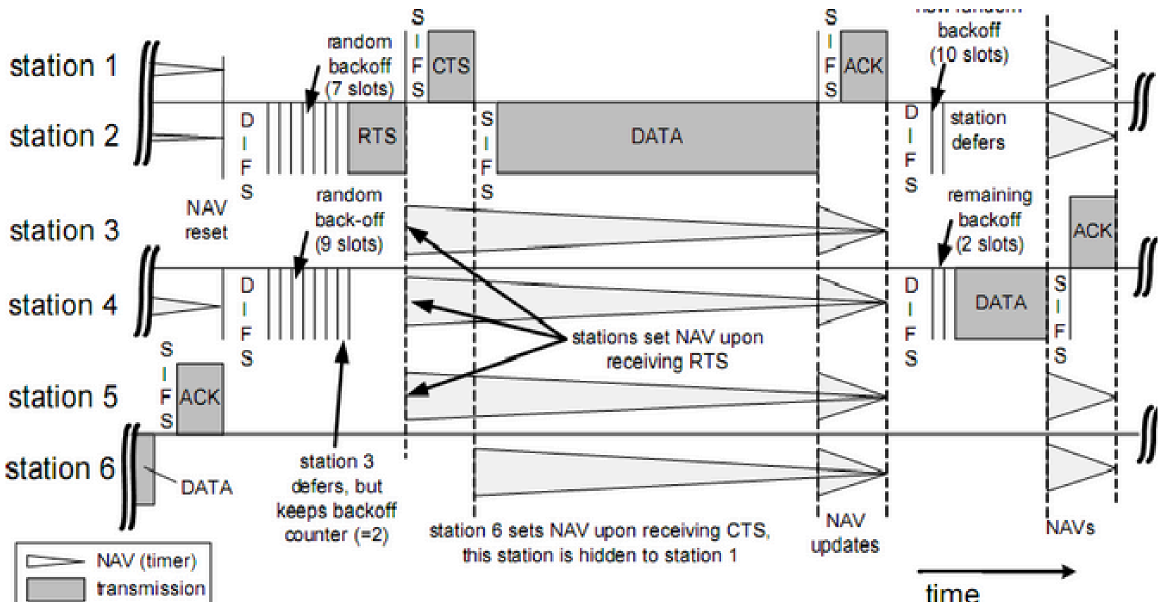
Hidden Stations and RTS/CTS

- Hidden Stations problem arises when a station is able to successfully receive frames from two different stations but the two stations cannot detect each other.
- To reduce throughput reduction owing to this problem, 802.11 specifies exchange of Request-to-Send/Clear-to-Send (**RTS/CTS**) frames as an option made by transmitting station.
- Consecutive frames in the sequence of RTS, CTS, data and ACK are spaced by an SIFS duration for transceiver turn-around.

Hidden Stations and RTS/CTS

- The RTS/CTS carry in its duration fields the information, how long the sequence RTS, CTS, Data, ACK will take.
- Hence, other stations close to the transmitting station and hidden stations close to the receiving station will not start any transmissions; their NAV timer is set.
- A hidden station close to receiving station might not receive the RTS due to the large distance, but will in most cases receive the CTS frame

Hidden Stations and RTS/CTS



- Station 6 cannot detect the RTS from station 2, but can hear the CTS of station 1. Although station 6 is hidden to station 2, it refrains from channel access because of NAV.
- Note that SIFS is shorter than DIFS, which always gives CTS and ACK the highest priority for access to the channel

Synchronization and Cell Search

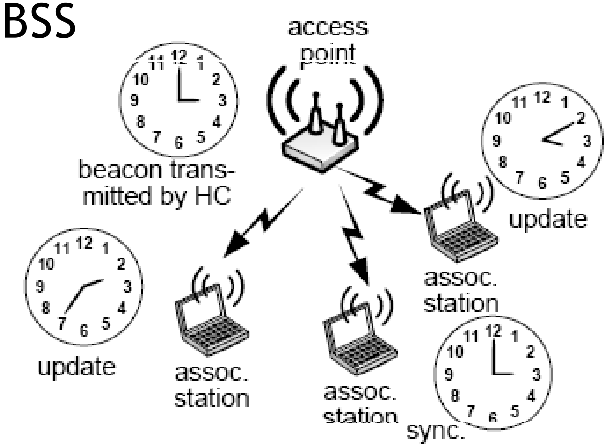
- All stations within a single BSS are synchronized to a common clock via the **Timing Synchronization Function (TSF).**
- To synchronize the stations, a management frame, the beacon, is used.
- The time when the next beacon frame arrive is called **Target Beacon Transmission Time (TBTT).** The beacons are transmitted periodically.

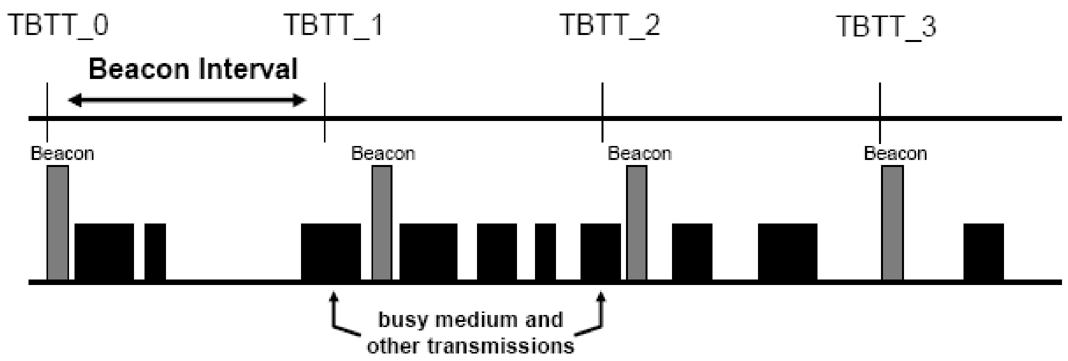
Synchronization and Cell Search

- TSF’s function
  - Support various PHY’s that require synchronization
  - Management function such as joining a BSS and saving power through sleep modes

Synchronization and Cell Search

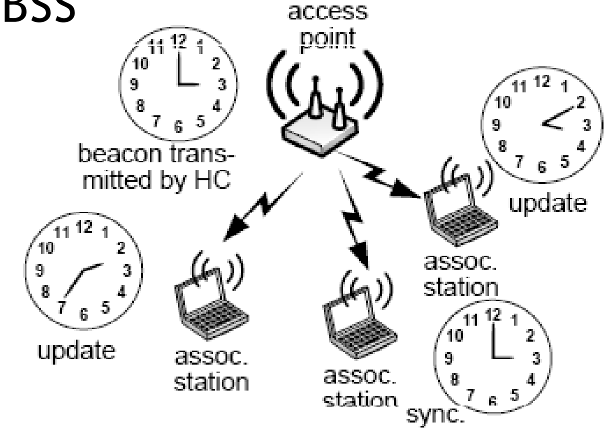
BSS

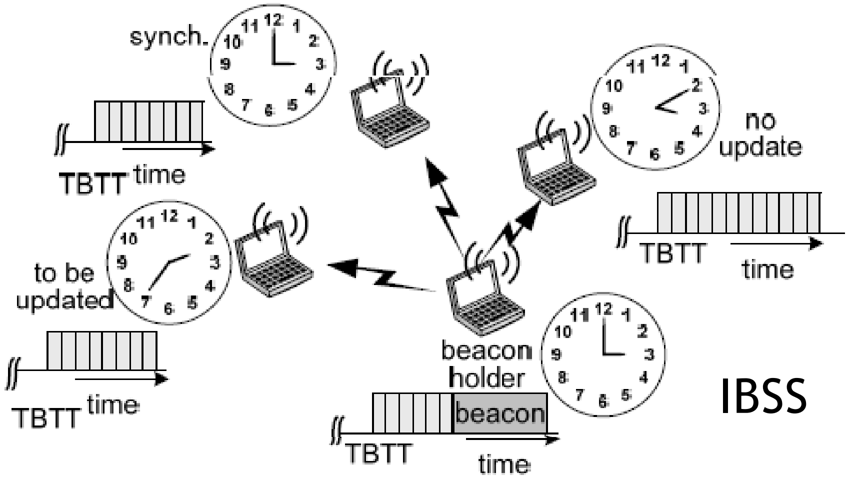




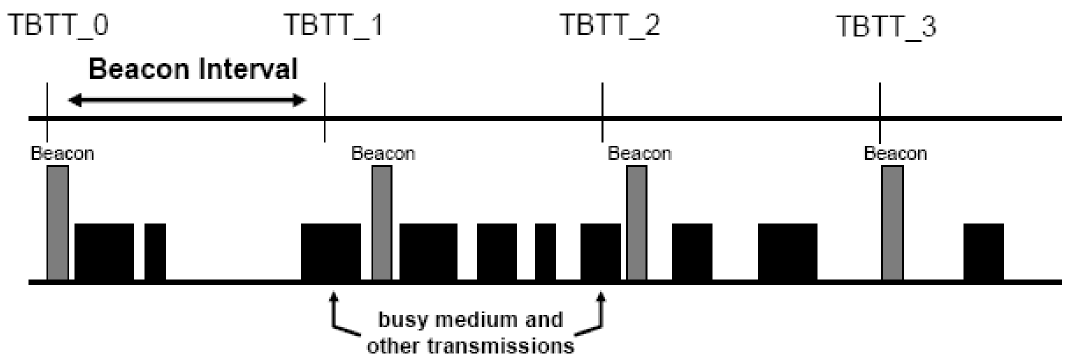
Synchronization and Cell Search

BSS





IBSS



Synchronization and Cell Search

- In BSS and IBSS, the synchronization is maintained by broadcasting the TSF timer in the beacon.
- In (infrastructure) BSS, only the AP generates beacons.
- If the channel has been idle for at least PIFS before TBTT, the AP transmits the beacon at TBTT, otherwise the beacon is transmitted PIFS <u>after the current transmission</u>.
- All stations associated to this AP update their local timers.

Synchronization and Cell Search

- In IBSS, the TSF is distributed over all stations. They all take part in the generation of beacons. The beacon generation is distributed using a mechanism similar to the backoff mechanism.
- Stations stop attempting to transmit a beacon when they receive a beacon from another station of the IBSS. However, beacons transmitted in contention may collide. Collided beacon is not retransmitted.
- Station updates its local timer only if the received value in beacon earlier then its current value.
- This distributed synchronization function results in the fact that the complete IBSS will synchronize to the fastest running clock.

Synchronization and Cell Search

- The beacon delivers other parameters related to the and radio regulations: for example
  - Basic Service Set Identification (BSSID),
  - Beacon Interval (BI),
  - PHY depending parameters,
  - duration of the CFP,
  - available channels,
  - power limits, etc.
- Depending on the type of BSS, not all information may be contained in the beacon.
- In infrastructure BSS, the AP uses the beacon for instructions to its associated stations and announcements of future transmissions.
- Knowledge about TBTT allows associated stations to operate in power save mode.
- Traffic indication messages conveyed together with each beacon

inform particular stations an buffered MSDUs and an upcoming multicast/broadcast messages.

Scanning Procedures in WLAN 802.11

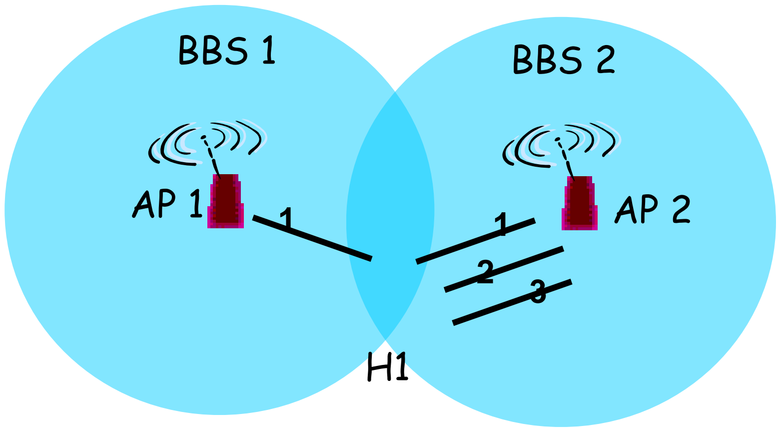
- IEEE 802.11 defines two scanning modes:
  - Passive
  - Active
- The decision on which scanning mode is to be applied is taken by the **Station Management Entity (SME).**
- The SME is a layer-independent entity that may be viewed as residing in a separate management plane.
- The exact functions of the SME are not specified in the standard.
- The SME executes actions related to general system management. It directs the MLME to perform either passive or active scanning.

Scanning Procedures in WLAN 802.11

host must *associate* with an AP

- scans channels, listening for *beacon frames* containing AP's name (SSID) and MAC address
- selects AP to associate with
- may perform authentication
- will typically run DHCP to get IP address in AP's subnet

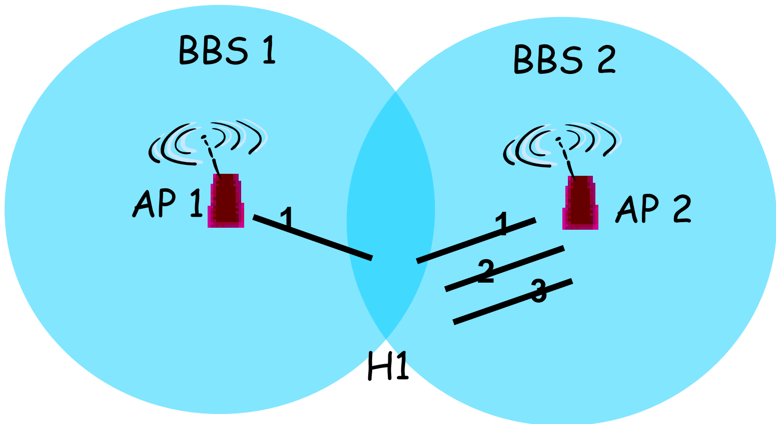
802.11: passive/active scanning

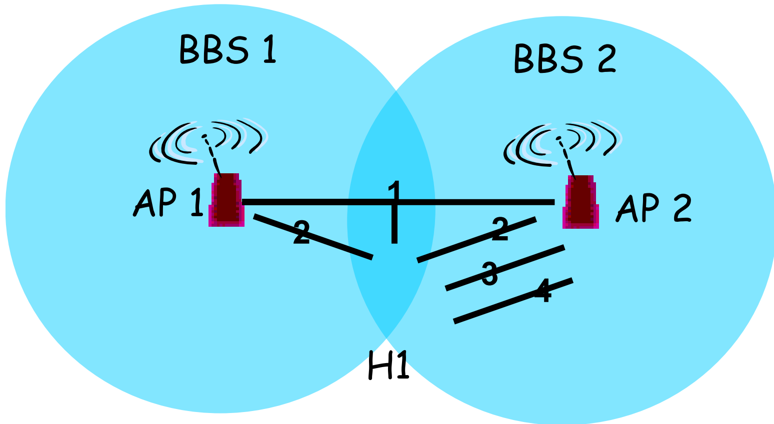


Passive Scanning:

1. beacon frames sent from APs
2. association Request frame sent: H1 to selected AP
3. association Response frame sent: H1 to selected AP

802.11: passive/active scanning





Passive Scanning:

1. beacon frames sent from APs
2. association Request frame sent: H1 to selected AP
3. association Response frame sent: H1 to selected AP

Active Scanning:

1. Probe Request frame broadcast from H1
2. Probes response frame sent from APs
3. Association Request frame sent: H1 to selected AP
4. Association Response frame sent: H1 to selected AP

Passive Scanning

- A station monitors each specified channel for a time span of at most **MaxChannelTime.**
- During the scanning, the station adds any received 802.11 beacon or probe response to its cached BSSID scan list.
- The scanning duration may vary, depending on the # of channels to be scanned, the BI, and the load of the system.

Active Scanning

- A station needs to generate specific request messages, so-called **probe frames**, then waiting for **probe response**. The probe response need to be acknowledged to ensure integrity of data delivery.
- The probe mechanism forces an AP to convey basically the same system information as done with the periodic beacon signal without unnecessary waiting times.

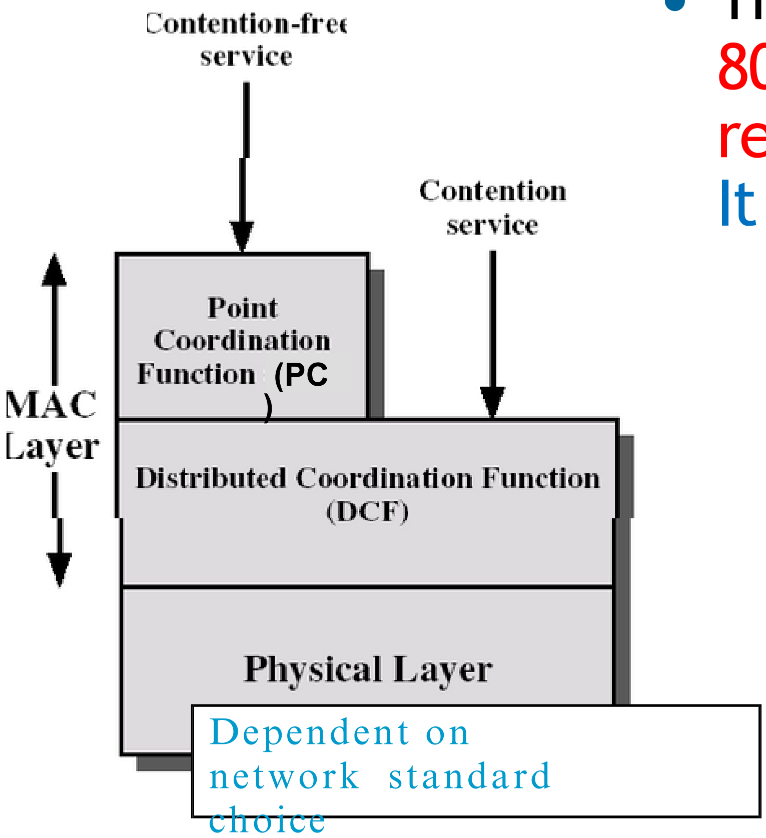
MEDIUM ACCESS CONTROL WITH SUPPORT FOR QUALITY-OF-SERVICE

IEEE 802.11
Point Coordination Function (PCF)

Medium Access Control with Support for Quality-of-Service

- IEEE 802.11 defines the **Point Coordination Function (PCF)** to let station have priority access to the radio channel, coordinated by a station called **Point Coordinator (PC)**.
- PC typically resides in the AP.
- The PCF has higher priority than the DCF, because the period during which the PCF is used is protected from the DCF access by the NAV.

Medium Access Control with Support Quality-of-Service

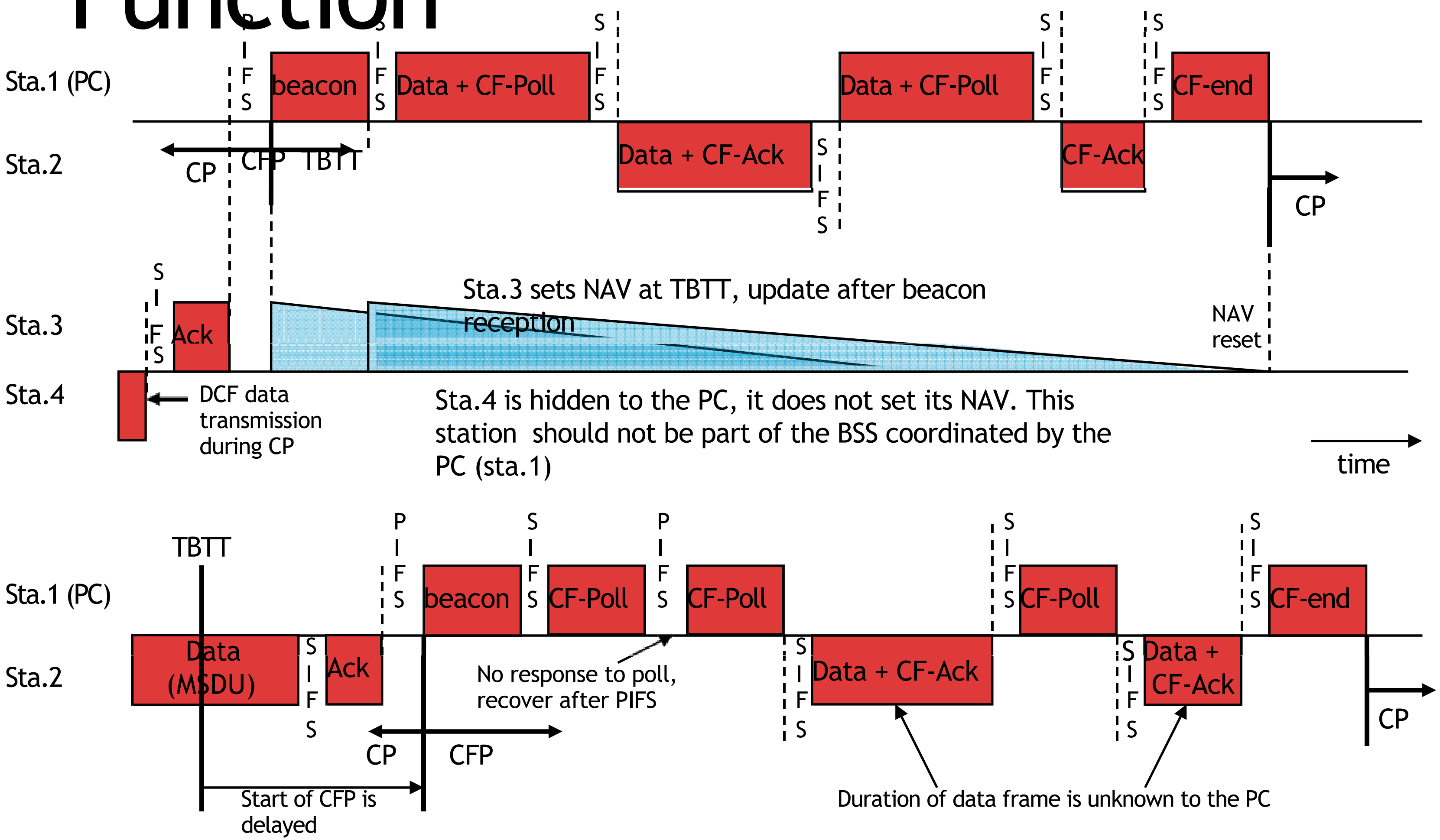


- The time during which 802.11 stations operate is divided into repeated periods, called superframes. It starts with a beacon, includes

  1. Contention Free Period (CFP),
  2. Contention period (CP).

- During the CFP, the PCF is used for accessing the channel, while the DCF is used during the CP.

Point Coordination Function



Point Coordination Function

- This previous figure is an example for the PCF operation. Station 1 is the PC and polls station 2.
- Station 3 detects the beacon frame and updates the NAV for the whole CFP.
- CFP may be delayed (lower part of the figure).
- PC has data pending for station 2
    - combine data and poll frame into the data frame.
- The PC continues polling other stations until the CFP expires. No idle period longer than PIFS occurs during CFP.
- A CF-End control frame is transmitted to signal the end of the CFP.

QoS Support with PCF

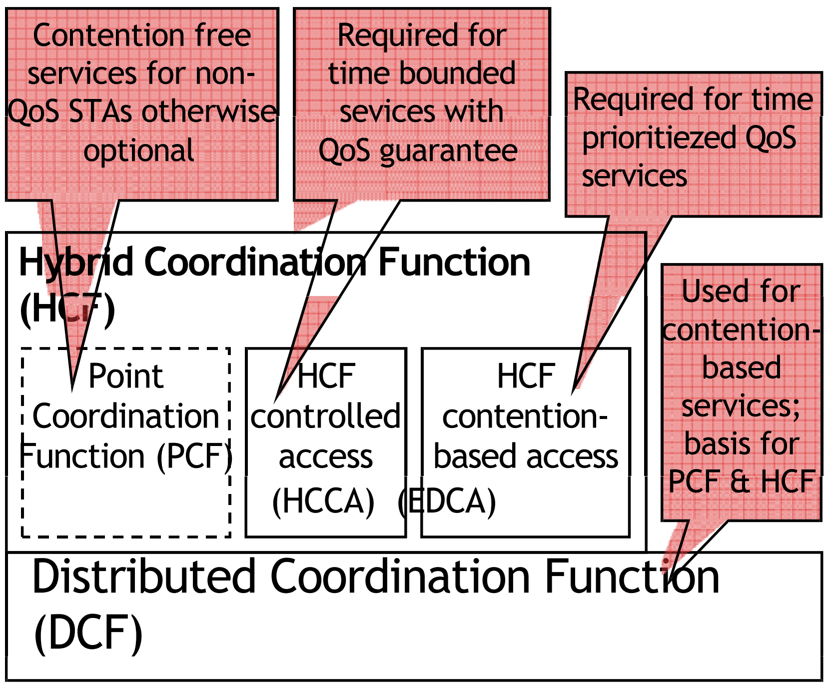
- The time the beacon is delayed from TBTT determines the delay of the transmission of time- bounded MSDUs that have to be delivered in the CFP.
- The duration of the MSDU delivery after polling is not under the control of the PC, which reduces the QoS provided to other stations that are polled during the rest of the CFP.

QoS Support Mechanisms of 802.11E

QoS Support Mechanisms of 802.11E

- The 802.11e extension introduces the Hybrid Coordination Function (HCF) for QoS support.
- The HCF defines 2 medium access mechanisms:
    - (i) the contention-based channel access (referred to as Enhanced Distributed Channel Access-EDCA);
      (ii) the controlled channel access, which includes polling (HCF Controlled Channel Access-HCCA).
- With 802.11e, there may still be the 2 phases of operation within a superframe:
    - Contention period (CP)
    - Contention Free Period (CFP)
- EDCA is used **only** in CP, while HCCA is used in **both** phases.

QoS Support Mechanisms of 802.11E



- Main elements of 802.11e MAC architecture in the context of 802.11.
- DCF is the basis for the contention-based access of the HCF and PCF.

QoS Support Mechanisms of 802.11E

- Concepts:
  - QBSS: a QoS supporting BSS, including an 802.11e-compliant HC.
  - HC (Hybrid Coordinator): the station that operates as the central coordinator for other stations, similar to PC in BSS.
- There are multiple backoff processes operating in parallel

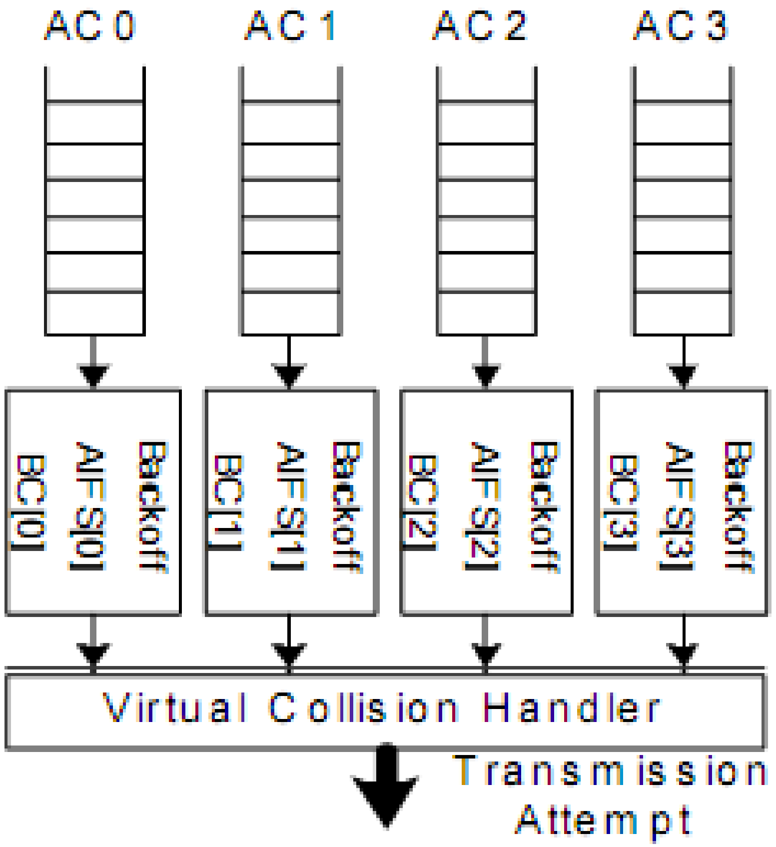
. , .

(Therefore, in the following we'll refer to backoff entities that attempt to deliver MSDUs, instead of stations)

Improvements of the 802.11 Legacy MAC

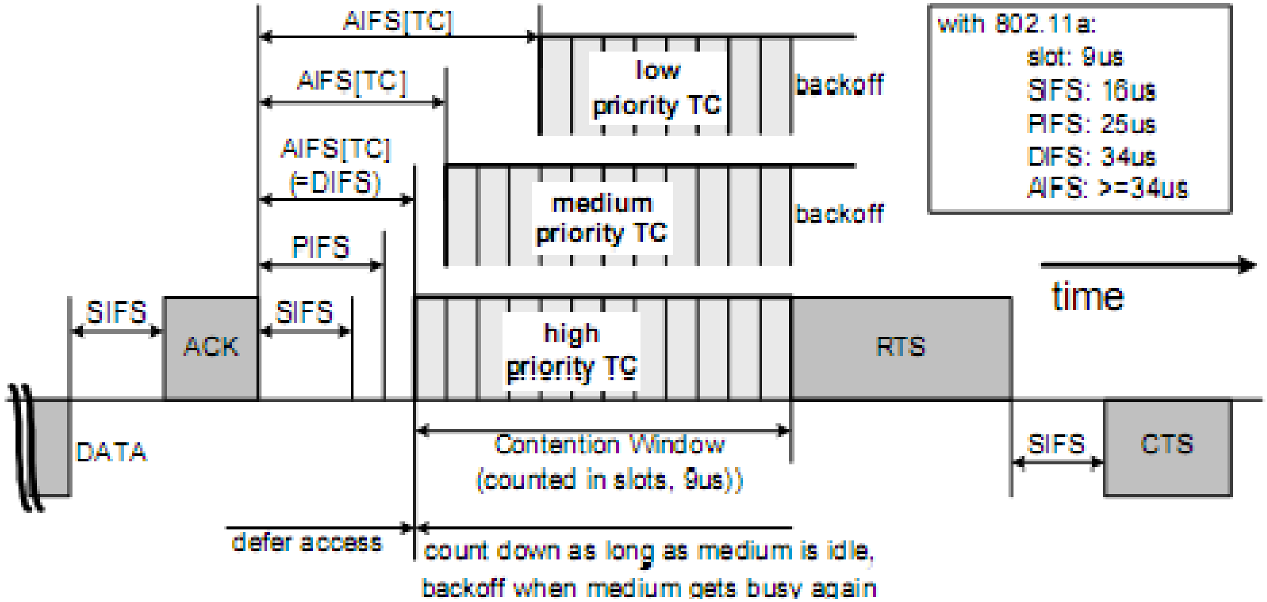
- A **TXOP (transmission opportunity)** is an interval of time, during which a backoff entity has the right to deliver MSDUs, is defined by its starting time and duration.
- TXOPs, which are obtained via the contention-based medium access, are referred to as **EDCA-TXOP**s, and limited by **TXOPlimit**
- **HCCA-TXOP**s (or polled TXOPs) are obtained by the HC via the controlled medium access, and protected by NAV.
- A time period in which the HC has control over the wireless medium is called **Controlled Access Phase (CAP)**

Contention-based Medium Access



- There are four Access Categories (AC), thus, <u>four backoff entities</u> exist in every 802.11e stations.
- ACs are labeled according to their target application: AC_VO (voice), AC_VI (video), AC_BE (best effort), AC_BK (background).
- Each backoff entity within a station <u>independently contends</u> for a TXOP.

Contention-based Medium Access



- After detecting the medium being idle for a duration of AIFS[AC] (Arbitration Interframe Space), backoff entity start down-counting. AIFS is calculated as follow:
  - AIFS[AC] = SIFS + AIFSN[AC]·aSlotTime >= 2
- Min and Max of the contention window are the other parameters dependent on AC.

Contention-based Medium Access

- The size of the CW in backoff stage i, after i-1 times unsuccessful transmission is:

$CWi[AC] = min[2^{I}(CWmin[AC]+1)-1, CWmax[AC]]$

- The collision probability increases with smaller CWmin[AC] if there are more than 1 backoff entity of the respective AC operating in the QBSS.
- The smaller CWmax[AC], the higher the medium access priority. However, a small CWmax[AC] may increase the collision probability.
- Besides retry counters similar to legacy 802.11, 802.11e also defines a maximum MSDU lifetime per AC to allow a frame remain in MAC.

Contention-based Medium Access

- Once a TXOP is obtained, a backoff entity may continue to deliver more than 1 MSDU consecutively during the same TXOP, which may take up to **TXOPlimit[AC]**. This concept is referred to as continuation of an EDCF-TXOP.
- Each AC has: **QSRC[AC], QLRC[AC]** (short and long retry counters). They are used for MAC frames of different lengths. Their default values are not given.
- When more than 1 entities in the same station reach 0 at the same time, a virtual collision occurs.
- The backoff entity with higher priority will transmit, whereas the others act as if a real collision occurred on the medium.

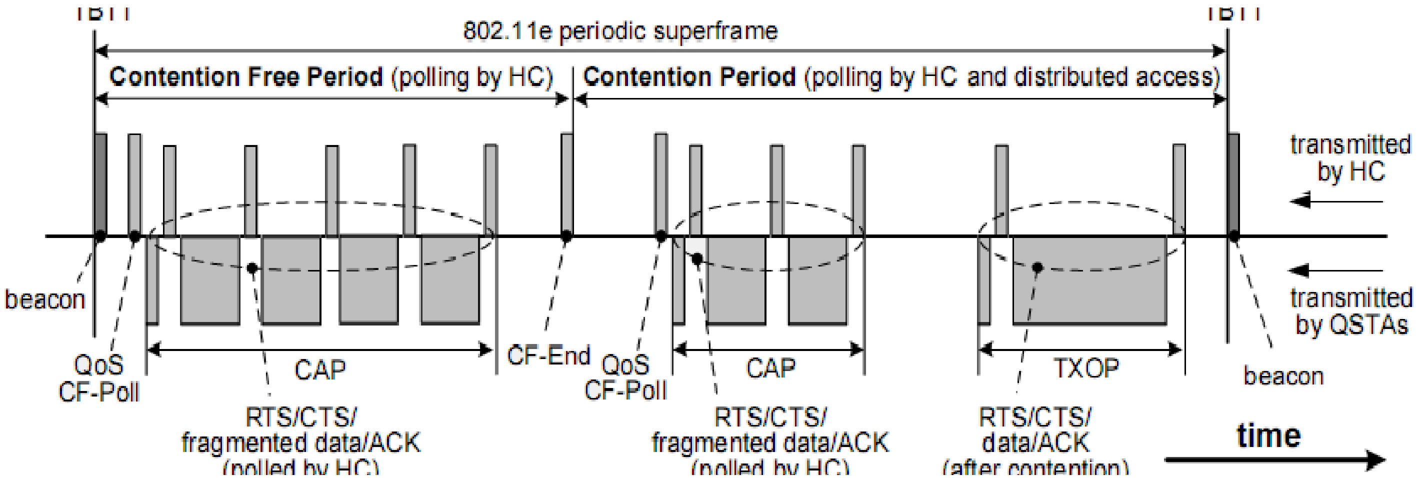
Controlled Medium Access

- The controlled medium access of the HCF, referred to as HCCA extends the EDCA access rules by allowing the highest medium access to the HC during both CFP and CP.
- HC may allocate TXOPs to itself to initiate MSDU deliveries whenever required, after detecting the medium idle for PIFS duration and without backoff.
- To give HC higher priority over legacy DCF and EDCA, AIFSN[AC] must be selected such that the earliest medium access for EDCA is DIFS for any AC.
- QoS CF-Poll from HC can be transmitted after a PIFS idle period, without backoff.

Controlled Medium Access

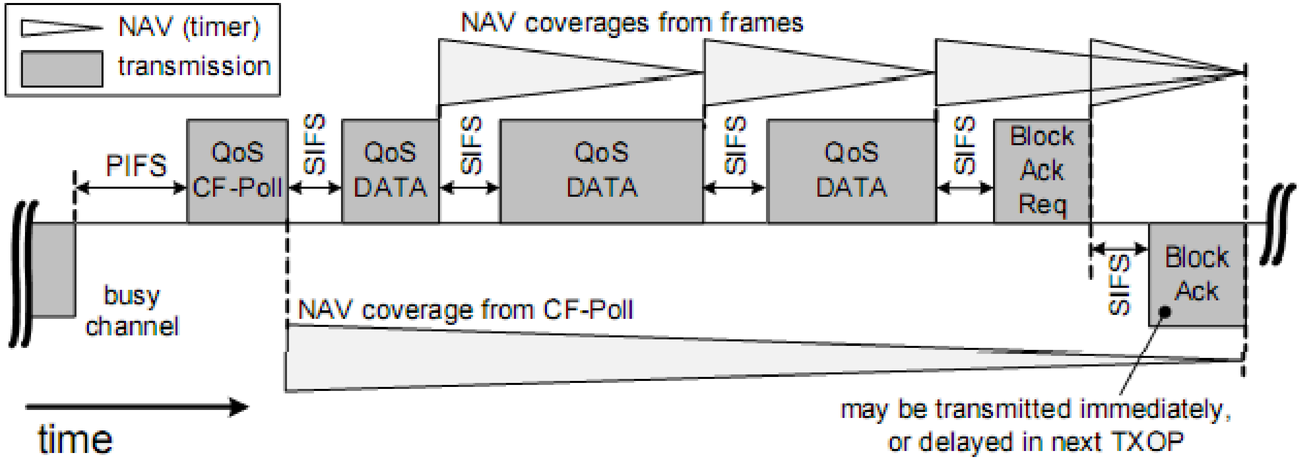
- During CFP, 802.11e backoff entities will not attempt to access the medium without being explicitly polled, hence, only HC can allocate TXOPs by transmitting QoS CF-Poll.
- During a polled TXOP, a polled station can transmit multiple frames, with SIFS between 2 consecutive frames as long as the entire exchange duration does not exceed TXOPlimit.

IEEE 802.11e Superframe



- The superframe starts with a beacon transmitted by HC.
- During CFP, the backoff entites only transmit upon being polled by HC.
- During the CP, HC may poll a station, which is different from PCF of the legacy 802.11

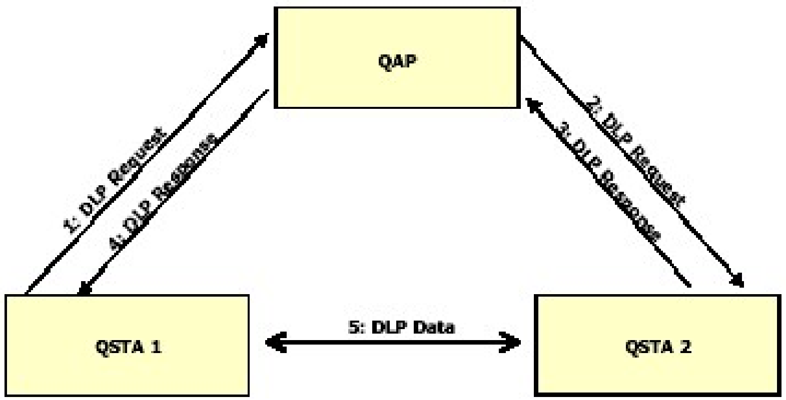
Block Acknowledgment



- Block acknowledgments allow a backoff entity to deliver a number of MSDUs being delivered consecutively during one TXOP and transmitted without individual ACK. € Throughput efficiency is improved.

Direct Link Protocol (DLP) for IEEE 802.11 WLAN

Direct Link Protocol (DLP)



Overview of DLP for setting up a direct link

- In the legacy 802.11, within an infrastructure-based BSS, AP relays all the frames. This wastes the channel capacity compared to the direct
.
- The direct communication in 802.11e is referred to as Direct Link (DiL).
- A set up procedure, the Direct Link Protocol (DLP) is defined to establish a DiL between 802.11e backoff entities.

802.11 Unicast vs. Broadcast

802.11 Unicast

- RTS/CTS
- ACK/Re-tx
- Exponential backoff
- Auto-rate adaptation

802.11 Broadcast

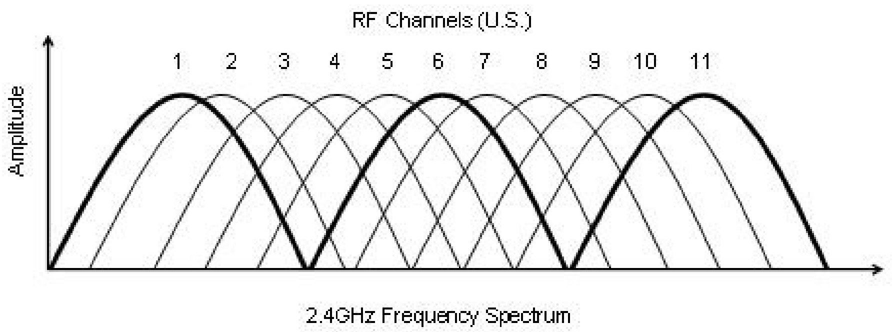
- No RTS/CTS
- No ACK/Re-tx
- No exponential backoff
- Single (low) rate

RADIO SPECTRUM MANAGEMENT

802.11: Channels

802.11b/g: 2.4GHz-2.485GHz spectrum divided into 11 channels at different frequencies

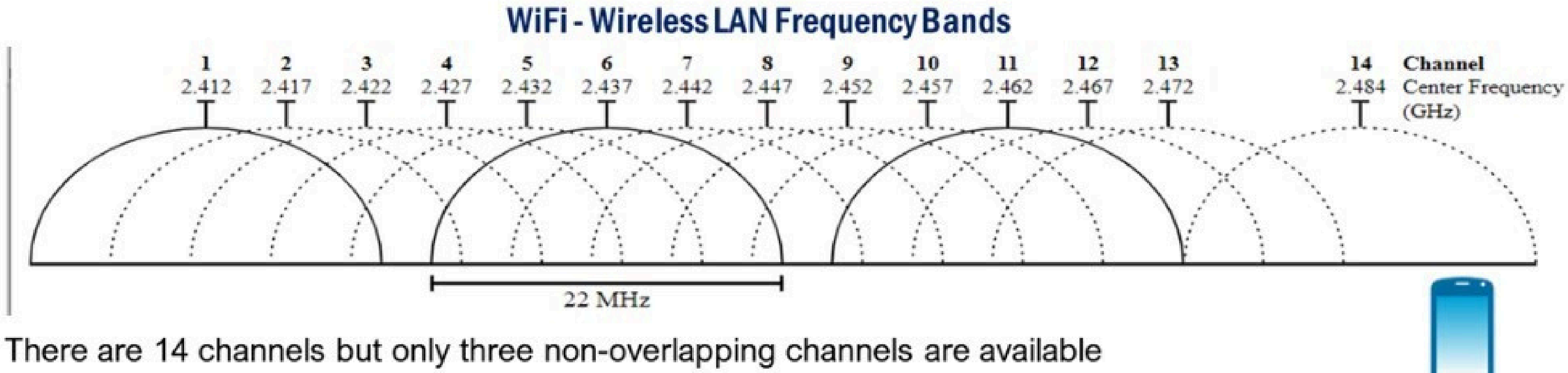
AP admin chooses frequency for AP
interference possible: channel can be same as that chosen by neighboring AP!

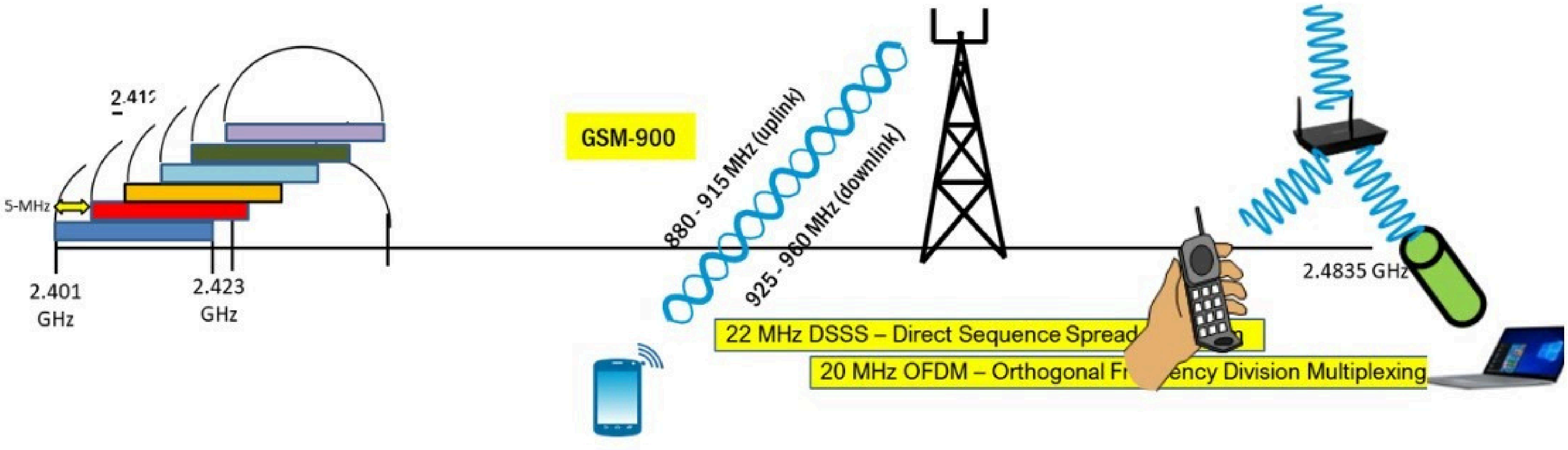


Only 3 non-overlapping channels!

802.11a: 8 non-overlapping channels

802.11: Channels





802.11: Channels